Chapter 6

Dental Biomaterials

"Nothing endures but change".

—Heraclitus, Greek Philosopher (540–480 BC)

This chapter discusses dental biomaterials from the perspective of the clinician trying to make a decision about, what material is best for his or her practice in order to provide optimum care for patients.

These decisions are often important because, in addition to clinical time, materials themselves are very expensive and the cost of premature failure of restoration and subsequent replacement is considerable. Most materials are brought to the marketplace with little or no clinical testing. Practicing dentists are basically doing the clinical testing for manufacturers. Understanding the basic nature and composition of dental materials can allow clinicians to make educated decisions related to new materials and avoid catastrophic outcomes (Fig. 6.1).

Contemporary dental practitioners are expected to practice **evidence-based dentistry.**[1–4] The evidence-based diagram (Fig. 6.2) shows that the top of the hierarchy is filtered information from meta-analyses and systematic reviews of published randomized controlled trials (RCTs). However, the reality is that it takes at least 10 years from the introduction of a new material before 5-year RCTs are published. Most currently published (2017) systematic reviews are of limited value to the clinician because the clinical trials included in the reviews are usually very weak and poorly designed.[5] Thus, in the absence of good 'evidence', it is essential that dentists understand the composition of commonly used dental materials and the functions of each component of those materials.

Dental materials have a number of important physical and mechanical properties that will be described in the next section. One of the most important properties is **technique sensitivity**. A material is described as technique sensitive when different operators get different outcomes while using that material. Materials that are low in technique sensitivity are desirable, and manufacturers strive to develop materials that are not technique sensitive. A good example of this is seen with dental adhesives. Dental adhesives have evolved over many years, and as many as 7 generations have been described. For many years, the 4th generation of dental adhesives was identified as the best system, but dentists have had variable results with it, as it required three separate steps (etch, prime, bond) and involved the vague concept of 'moist' dentin. These materials were indeed 'technique sensitive'. Subsequent generations attempted to simplify the bonding procedure and make it less technique sensitive, but in fact made it less effective. Dentists, who understood the 4th generation of bonding agents and who were meticulous in their technique, achieved excellent clinical results. Those who did not understand these materials and/or were less meticulous in their technique got less successful outcomes.

One material that is very low in technique sensitivity is silver amalgam (Fig. 6.3). A short time after placement of a silver amalgam restoration, oral fluids penetrate the microscopic space between the amalgam and tooth structure.[6] This process, known as **percolation**, results in the deposition of corrosion products and effectively seals the interface between the amalgam and tooth. This process of 'self-sealing' occurs irrespective of the skill and knowledge of the operator, which makes silver amalgam one of the least technique-sensitive materials.

The discussion on technique sensitivity leads to the concept that there are two parts in the successful use of dental materials: **material selection** and **material manipulation.** Dentists must select the most appropriate restorative material based on knowledge of the disease process, knowledge of materials appropriate for dealing with the dental defect and knowledge of expected outcomes of the proposed intervention. Appropriate material manipulation of a selected material requires knowledge about the material and meticulous attention to details such as isolation, bonding technique, incremental build up, curing time, matrix selection and many other variables. Thus, both material selection and proper manipulation of the selected material are critical to clinical success.



Figure 6.1 Fractured porcelain veneer.

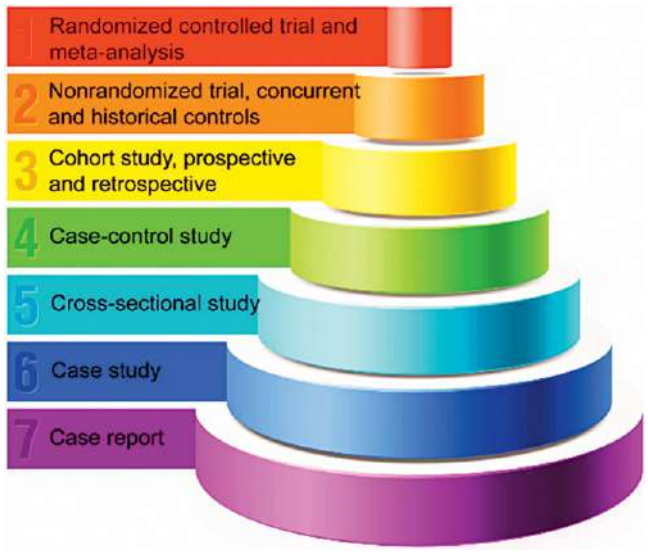


Figure 6.2 Evidence-based medicine pyramid.

Clinical Notes

I. Understanding the composition and nature of dental materials is important to obtain optimum clinical outcomes.
II. Materials are brought to the market with almost no clinical testing.
III. Premature clinical failures of dental restorations are extremely costly to dental practices.
IV. The evidence base for restorative materials is not robust.
V. The relative technique sensitivity of a material is an important attribute.
VI. Both material selection and material manipulation are important.



Figure 6.3 Amalgam restorations.

Physical, Mechanical and Optical Properties

A number of important physical and mechanical properties of dental materials can be identified. This section will discuss some of these properties and illustrate how they might be important to the long-term clinical success of a restoration. This section is not meant to be an exhaustive description of all the physical and mechanical properties that can be evaluated in dental materials. Rather, it is intended to be a brief description of several properties of dental materials that might be critical for clinical success.

The following is an incomplete list and description of physical and mechanical properties of dental materials and some examples of why they are important.

Dimensional Change

Dimensional change is the percent shrinkage or expansion of a material, which usually occurs as a result of the setting reaction.

One example is with composite resin. Basically, composite resin materials consist of inert filler particles in a matrix of resin, usually bisphenol A glycidyl methacrylate (BIS-GMA). The resin material undergoes a polymerization reaction as it sets, with individual monomers joining to become long-chain polymers, which then cross-link to form the polymer matrix. During the process, the material undergoes polymerization shrinkage. With contemporary hybrid composite resin materials, the shrinkage is between 2.4% and 2.8%. If not accounted for in the placement technique, this amount of shrinkage can lead to clinically significant contraction gaps between the margin of the preparation and the composite resin restoration. This can result in microleakage, postoperative sensitivity and recurrent caries.

Thermal Coefficient of Expansion (COE)

The coefficient of expansion (COE) is defined as the amount a material expands per unit length if heated 1 degree higher.

Waxes have the highest COE of all dental materials. When the lost wax process is used, wax patterns should be invested as soon as possible after margination to ensure

Table 6.1 Linear thermal coefficient of expansion of dental materials in the temperature range 20°C to 50°C

Materials	Coefficient ($\times 10^6$/°C)
Human teeth	8–15
Ceramics	8–14
Glass ionomer base	10–11
Gold alloys	12–15
Dental amalgam	22–28
Composites	25–68
Unfilled acrylics and sealants	70–100
Inlay wax	300–1000

that there is no distortion of the wax pattern due to normal fluctuations in room temperature (Table 6.1).

This property can be very important when dealing with materials that have cores with veneering materials. An example is metal-ceramic crowns. The feldspathic porcelain veneer, which provides excellent esthetic result with such restorations, is very brittle and must be supported with a rigid material (metal alloy). When fabricating metal-ceramic restorations, the crowns must be heated to a high temperature several times to sinter the veneering ceramic. Then the units must be cooled to bench temperature. If the COEs of the metal alloy and veneering porcelains are not closely matched, stresses can build up in the veneering porcelain that result in cracking or fracture of the ceramic. This can occur at the time of fabrication or later in clinical service. Generally, the COE of the metal alloy is reduced with the addition of platinum, and the COE of the porcelain is increased with the addition of sodium. Whatever the mechanism used, it is clear that the COEs of both the core and veneering material must be closely matched (Fig. 6.4).

It is often thought that the COEs of restorative materials must be matched with the COE of tooth structure. If they are not matched, the ingestion of hot beverages (e.g. coffee) and cold foods and beverages (e.g. ice cream) could result in uneven expansion and contraction of the restoration and tooth, resulting in microleakage. The reality is foods and beverages do not surround the tooth/restoration complex for a long enough time frame to effect any significant change in the temperature of the tooth restoration.



Figure 6.4 Metal-ceramic restorations.

Percolation

The process of fluids entering the microscopic space between a restorative material and the wall of the cavity preparation is called percolation.

This is not a beneficial process with materials such as composite resins, but works wonderfully with materials such as silver amalgam. When percolation occurs with amalgam, corrosion products build up at the tooth–restoration interface and effectively seal it, usually 6 weeks after placement of the amalgam. With modern high-copper amalgams with reduced gamma-2 phase, this phenomenon occurs but takes slightly longer to create an effective seal (Fig. 6.5; Table 6.2).

Thermal Conductivity

Thermal conductivity is the rate at which thermal changes are conducted through a material.

Materials with high thermal conductivity are good conductors of hot and cold, and could be a factor in postoperative sensitivity. This is the rationale for placing bases and/



Figure 6.5 Teeth numbers 12, 13, 14, 15 restored with amalgam restorations. Percolation = corrosion product = seal.

Table 6.2 Thermal conductivity of different materials used in dentistry

Materials	Thermal conductivity (cal/sec/cm^2[°C/cm])
Unfilled acrylics	0.0005
Zinc oxide–eugenol cement	0.0011
Human dentin	0.0015
Human enamel	0.0022
Composites	0.0025
Ceramic	0.0025
Zinc phosphate cement	0.0028
Dental amalgam	0.055
Gold alloys	0.710

or liners with cavity preparations that are close to the pulp. Bases and liners have low thermal conductivity and thus are efficient insulators.

There is a speculation by some that these bases and liners are not needed because dentin itself is a good insulator. However, often the thickness of dentin over the pulp may be minimal, so *it is usually recommended that the thickness of remaining dentin and the thickness of the base or liner should be at least 2 mm*.

It is interesting to note that amalgam is an excellent thermal conductor, whereas composite resin is an effective insulator; however, the incidence of postoperative sensitivity is significantly higher with composite resin restorations than with amalgam restorations. This demonstrates that the phenomenon of postoperative sensitivity is multifactorial and complex, and not due to thermal conductivity alone.

Layered zirconia restorations experience a rate of veneer chipping about 8 times that of metal-ceramic restorations. The etiology of this chipping is multifactorial, but one of the potential causes is the thermal conductivity of the zirconia core. The thermal conductivity is extremely low, and zirconia can be considered as an insulator. The consequence of this is that when a layered zirconia restoration is removed from the sintering oven, the core cools very slowly and the veneering ceramic cools past its glass transition temperature, while the core continues to cool and shrink. The net result of this interaction is that stress is built up in the veneering ceramic, which is later released via clinical chipping of the veneering material. New firing cycles involving slower cooling rates seem to have reduced the incidence of chipping in recent years.[7]

Creep

Creep is defined as time-dependent plastic deformation of a material under static load or constant stress.

This property is important with silver amalgam. Conventional low-copper amalgam materials have high creep values and exhibit poor marginal integrity over time, as the amalgam creeps into a thin layer and then breaks off creating marginal ditching. Contemporary high copper amalgams have low creep values and maintain good marginal integrity as a function of time.

Creep is one of the few properties that have a demonstrated ability to predict clinical performance. Amalgam materials with low creep values perform well clinically when compared with amalgams with high creep values.

Electrical Properties

The three major electrical properties in dentistry are galvanism, tarnish and corrosion.

I. **Galvanism** occurs whenever different metals are in contact with one another in the presence of an electrolyte (saliva).



Figure 6.6 The generation of electrical currents due to contact of dissimilar metals in an electrolyte is known as galvanism.

Clinically, this can occur when a gold crown is placed adjacent to an existing amalgam restoration or vice versa. Gold crowns adjacent to metal-ceramic crowns can have a similar result, as there are over 277 different commercially available porcelain-bonding alloys. The clinical symptoms resemble an 'electrical shock' or the presence of a metallic taste. These symptoms typically are self-limiting and are not considered serious problems (Fig. 6.6).

II. **Tarnish** is a surface phenomenon resulting in unesthetic discoloration of dental alloys, similar to tarnish of commonly used silverware (Fig. 6.7).

It does not result in physical breakdown of the alloy. It is important to note that the term **noble metal** connotes that the metal is resistant to tarnish in the oral cavity. The major noble metals used in dentistry are gold, platinum and palladium. Others are rhodium, ruthenium, iridium and osmium. Alloys should contain a minimum 45% of noble metal to prevent intraoral tarnishing. Chromium is a metal that prevents tarnish in base metal alloys. One metal that is precious but not noble is silver.

III. **Corrosion** is chemical or electrochemical dissolution of metals in oral fluids.



Figure 6.7 Tarnish of silverware. Dental alloys may undergo the same discoloration in the oral environment.



Figure 6.8 Example of corrosion due to electrochemical dissolution in water.

The varying pHs in the oral environment create the possibility of corrosion, although in fact it is a rare occurrence. Corrosion of prefabricated metal endodontic dowels with fractured endodontically treated teeth has been seen occasionally. In these cases, it is believed that fracture of the endodontically treated teeth occurred first, allowing contact of oral fluids with the post-resulting in corrosion. The other possibility is that the corrosion occurred first followed by fracture due to the production of corrosion products (Fig. 6.8).

Solubility

Solubility is a relative tendency to dissolve in oral fluids.

This was a major problem with the first direct esthetic restorative material, *silicate cement*. This cement was a powder liquid system, the powder being ground aluminosilicate glass and the liquid phosphoric acid. When mixed, it formed a paste that was inserted into the prepared cavity and when set and finished provided a restoration suitable for class III restorations that was expected to provide only 3– 5 years of service. The reason for this short time of service was that silicate cements were soluble in oral fluids. An unexpected benefit of this problem was that as the aluminosilicate glass dissolved, it released high concentrations of fluoride. When the partially dissolved restoration was removed for replacement, recurrent caries was never found due to this high level of fluoride release (Fig. 6.9).

Conventional non-resin-based dental cements had variable levels of solubility; for example zinc-oxide eugenol (ZOE) cements that are often used for provisional restorations are quite soluble in oral fluids. From a practical standpoint, this means that *provisional restorations should not be cemented for more than 6 weeks if ZOE is used*. At that time, the provisional restorations should be removed, cleaned and recemented.



Figure 6.9 Class III cavity restored with silicate cement after 3 years. Notice the poor resistance to solubility and colour change. No evidence of recurrent carries.

The relative solubility of conventional non-resin cements made it necessary to provide precisely fitting definitive restorations. Cast gold restorations provide an extremely long time of service cemented with zinc-phosphate cement, providing they have precision of fit. Poorly fitting cast gold restorations provide a short-term service, as they are simply chunks of metal in a sea of cement.

Solubility of dental cements is typically measured with a 24-h in vitro solubility test where a sample of cement is immersed in a solute for 24 h and the amount of dissolved cement is calculated.[8] Long-term (6 months) in vivo solubility tests are preferred. These types of studies indicate that there are small differences between cements in long-term solubility, but, importantly, find there are significant differences among patients in regard to solubility. Some patients dissolve considerable amounts of cement, while other patients demonstrate minimal solubility. The causes of this phenomenon are not known.

Glass ionomer cements (GIC) perform poorly in 24-h in vitro solubility testing because they are quite soluble in the first 24 h after mixing. However, they perform very well in 6-month in vivo testing, and in fact are the best of the conventional non-resin cements in that parameter. This means they must be protected from saliva immediately after cementation using excellent isolation and coating with petroleum jelly or varnish.

Optical Properties

Optical properties of bulk materials include interactions with visible light that involve *reflection*, *refraction*, *absorption* (*and fluorescence*) *or transmission*.

The radiation typically involves different intensities for different wavelengths (or energies) over the range of interest (spectrum). Any of these interactive events can be measured using a relative scale or an absolute scale. When the electromagnetic radiation is a visible light, the amount of reflection can be measured in relative terms as gloss or in absolute terms as percent reflection. The visible light absorption can be measured in absolute terms as percent

absorption (or transmission) for every wavelength (in the visible spectrum).

Colour is a perception by an observer of the distribution of wavelengths. The same colour sensation may be produced by different absorption spectra (metamerism). An individual's eye is capable of sensing dominant wavelength, luminous reflectance (intensity) and excitation purity. Variations among individuals' abilities to sense these characteristics give rise to varying perceptions of colour.

The quality of colour is measured by the *Commission Internationale de l'Eclairage system* as tristimulus values and reported as colour differences (ΔL*, Δa* and Δb*) compared with standard conditions.

Radiation of another wavelength may be preferentially absorbed (e.g. X-rays). Composites that contain lithium, barium, strontium or other good X-ray absorbers may appear radiopaque (radiodense) in dental radiographs. The materials that are good absorbers (for whatever form of radiation) are described as opaque.

The appearance of a dental restoration is a combination of events of surface reflection, absorption and internal scattering. The scattering simply may deflect the path of the radiation during transmission (refraction), or it may reflect the radiation internally from varying depths back out of a solid to the observer (translucency). Enamel naturally displays a high degree of translucency; translucency is a desirable characteristic for restorative materials attempting to mimic enamel. A tooth that is isolated and kept from any contact with oral moisture (saliva) soon develops a transient whiter appearance. *Refractive index is the angle of changed path for a standard wavelength of light energy under standard conditions* (Fig. 6.10).

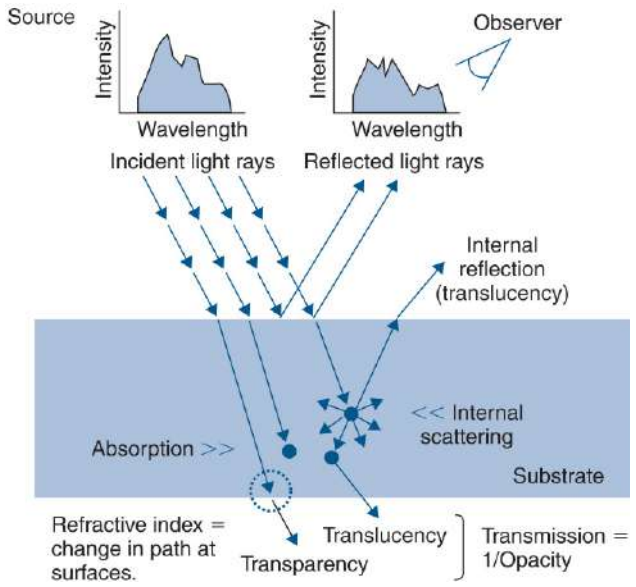


Figure 6.10 Schematic summary of interactions of electromagnetic radiation with materials. The colour perceived by the observer is the result of several interactions between substrate and incoming radiation producing reflection, internal scattering, absorption, fluorescence and transmission.

Compressive Strength, Tensile Strength, Diametral Tensile Strength, Flexural Strength and Fracture Toughness

All of these are mechanical properties of materials that are tested utilizing non-clinical samples and thus are only useful when comparing the tested material to a known standard. Compressive strength tests use cylindrical samples. Tensile tests can be done with certain materials such as wires but cannot be used with materials such as dental cements. Diametral tensile strength tests use discs and apply the force to an edge. As these tests do not test the material in the mode in which it is used, data obtained from the tests are of little value in predicting clinical performance but may be useful in comparing a new product to a standard (Fig. 6.11).

Flexural strength testing has become popular for evaluating new ceramic materials. Many of the newer ceramics such as lithium disilicate and monolithic zirconia have vastly improved flexural strength compared to feldspathic porcelain or leucite reinforced ceramic. However, that data in itself does not automatically ensure improved clinical performance, because ceramic crowns do not fracture due to lack of strength. They fracture because of inherent defects called *Griffith flaws*, which are incorporated into the restorations during fabrication. These defects propagate under occlusal stress and due to static fatigue, and eventually result in catastrophic failure. These newer materials may well provide improved clinical performance because the techniques used in fabrication (pressing and machining) result in fewer inherent defects, and the phase transformation capability of zirconia prevents defects from propagating after initiation.

Fracture toughness determination may be one of the most valuable in vitro tests because it evaluates failure initiating from a predetermined defect. It is possible that materials with improved fracture toughness may show improved clinical performance.

Elastic Modulus

Elastic modulus is a measure of the relative stiffness or rigidity of a material (Fig. 6.12). The elastic modulus is the ratio of stress over strain and is the slope of the straight-line portion of a stress–strain diagram. It is a measure of how the material will deform when placed under stress. A material with a high elastic modulus will be *stiff*, and a material with a low elastic modulus will be *flexible*.

Ceramic materials have a very high elastic modulus and are very brittle. These materials are frequently bonded to a core material to support the ceramic veneer and prevent fracture. It is critical that the core material have an elastic modulus equal to or higher than that of the ceramic material, because if the core material flexes, the ceramic will fracture.

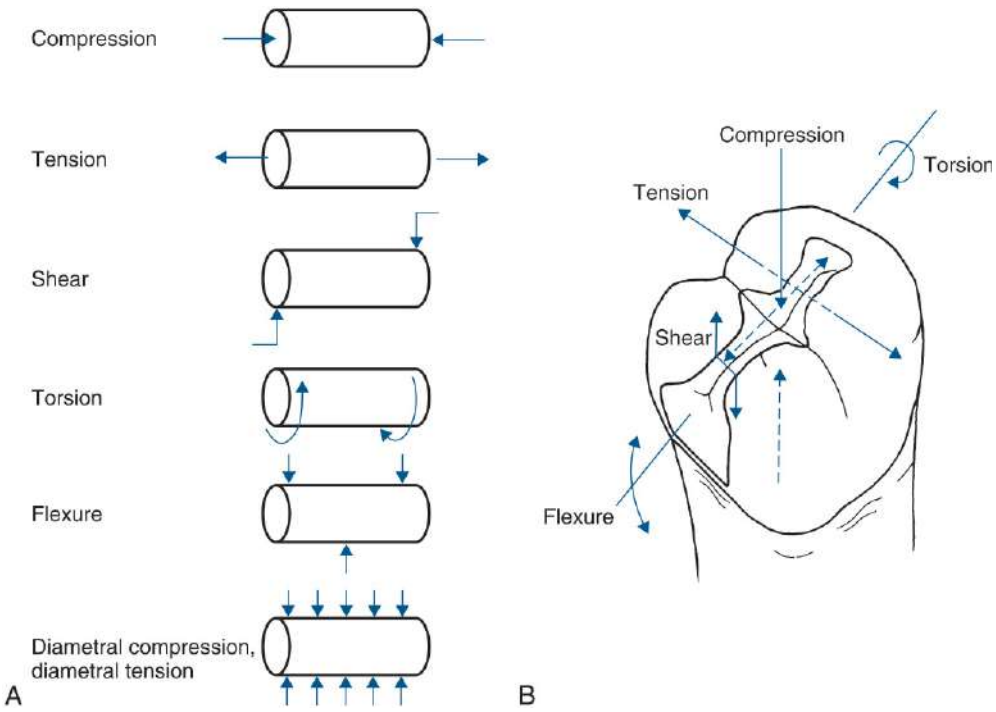


Figure 6.11 Examples of directions of loading. (A) Uniaxial loading of cylinder. (B) Uniaxial loading of a mesioocclusal amalgam restoration.

Another area where the modulus of elasticity of a material is important is the restoration of non-carious cervical lesions (NCCLs). One potential etiology of NCCLs is tooth flexure leading to cervical breakdown. Traditionally, materials with a low elastic modulus such as micro-filled composite resins and resin-modified glass ionomers have been more successful in restoring NCCLs than stiff materials such as hybrid composite resins.[9] Contemporary adhesive systems are now highly filled; and because they have a low elastic modulus, can act as stress breakers during tooth flexure, permitting successful restoration with stiffer composite resins.

Load-to-Failure Testing Versus Fatigue Testing

The traditional approach to test the strength of dental restorations is load-to-failure testing. With load-to-failure testing, a restoration is fabricated and cemented/bonded to the prepared tooth or die and then loaded with a testing machine and the forces increased until the restoration fractures. If the load at which the restorations fail exceeds the maximum bite force in the area of the restoration, then it is deemed that materials will provide excellent clinical performance. Data generated from load-to-failure testing can be very misleading and do not provide predictive information on clinical performance.

Several problems with load-to-failure testing can be identified. Clinically, teeth and restorations generally fail by defect propagation. Load-to-failure testing does not take this into account. *The loads at which failure occurs with load-to-failure testing are much higher than the loads at which clinical failure occurs.*[10] The failures with load-to-failure testing initiate at the occlusal surface of the restoration being tested while clinically restorations fail by defect propagation from defects at the tooth cement surface. Finally, the fracture pattern of failed restorations with load-to-failure testing results in the restoration fracturing in multiple fragments. Clinically, failure usually occurs with the restoration breaking into two pieces.

One major argument against load-to-failure testing is that restorations do not fail because the patient bites down with incredible force on a restoration, but *rather they fail by defect propagation due to low occlusal forces over a long period of time*. In order to make in vitro testing more clinically related, fatigue testing using cyclic loading machines has become increasingly popular (Fig. 6.13).

Fatigue testing using cyclic loading is considerably more relevant than load-to-failure testing because it much more closely mimics the clinical situation. Fortunately, many major research organizations have made considerable investment in cyclic loading machines and more fatigue studies are appearing in the peer-reviewed literature. It is important to note that dental fatigue testing is in its infancy and many factors need to be considered when attempting to bring the study environment closer to the clinical situation.

Some of the variables that need to be considered include *loading force*, *number of cycles*, *loading frequency* and *antagonist material*. Whether to include a simulated periodontal ligament is an important consideration. Whether to fatigue

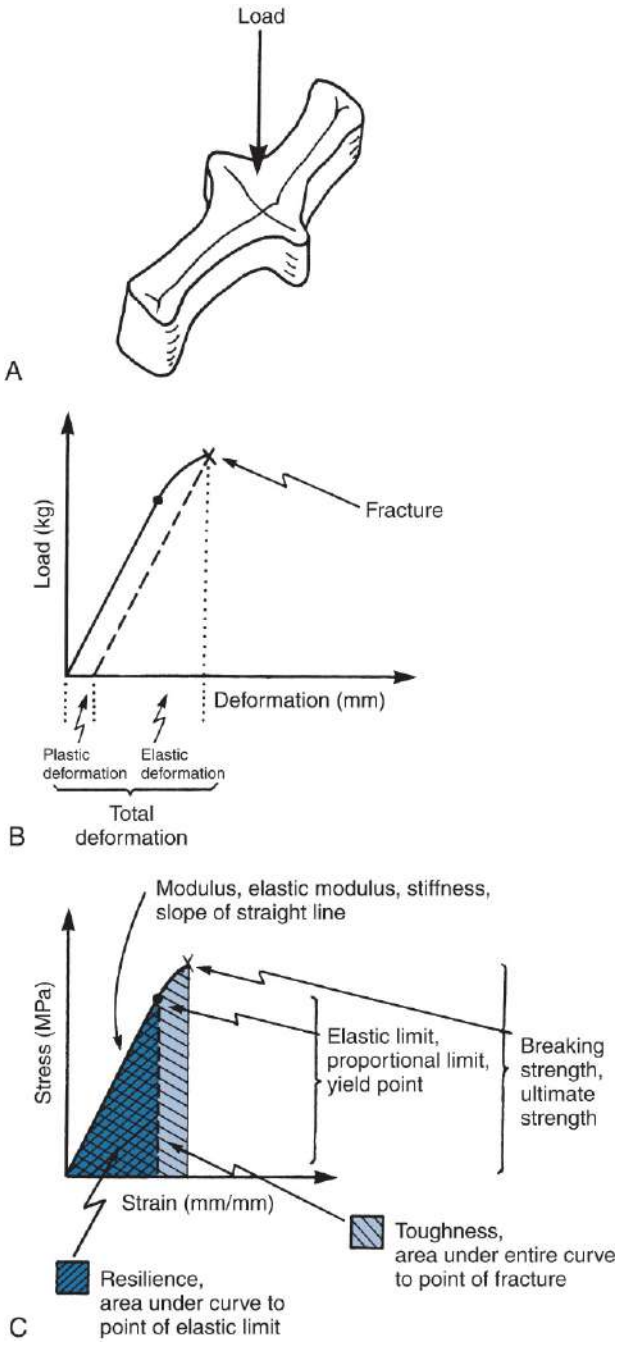


Figure 6.12 Schematic summary of mechanical properties with respect to amalgam restoration in function. (A) Occlusal loading of Class I amalgam restoration. (B) Load/deformation curve describing behaviour of amalgam. (C) Normalization of load/deformation curve to stress-strain curve with important characteristics of curve indicated. (Mechanical responses depend on temperature and strain rate involved.)

under water or dry is also a consideration. With ceramic materials, it is critical to fatigue underwater, as it has been proven that ceramic materials fail at half the load under water as in air. This is because of the presence of static fatigue with ceramic materials underwater or saliva. The use of thermocycling, the type of abutment or die material, the



Figure 6.13 Eight-chamber chewing simulator.

use of vertical and lateral movements and the use of anatomic versus flat samples are all important considerations when designing a fatigue study.

The bottom line is that load-to-failure studies have no clinical relevance whatsoever and should not be published. While recognizing that designing appropriate fatigue studies is challenging, it is clear that these types of studies are going to play an increasingly important role in dental materials research.

Wettability

Wettability is a measure of the affinity of a liquid for a solid as indicated by the spreading of a sessile drop (Fig. 6.14).

This might be the best explained by the state of a raindrop on a freshly waxed car. It will bead up, indicating a high contact angle and low wettability. Similarly, placing a drop on air abraded versus polished metal will result in a spreading drop (low contact angle; good wettability) on the former and a beaded drop (high contact angle; poor wettability) on the latter.

Early versions of additional reaction silicone impression material were extremely hydrophobic and difficult to pour. Modern elastomeric impression materials contain nonionic surfactants, which render them more hydrophilic and considerably easier to pour because the surface is more wettable and forms a lower contact angle with liquids.

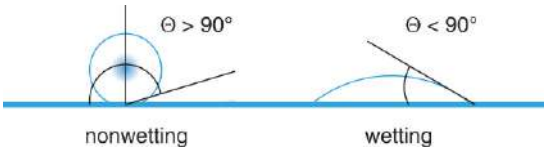


Figure 6.14 Schematic drawing of the contact angle determining wettability. Contact angles less than 90 degrees are considered to be wetting and hydrophilic. Contact angles larger than 90 degrees are considered nonwetting and hydrophobic.

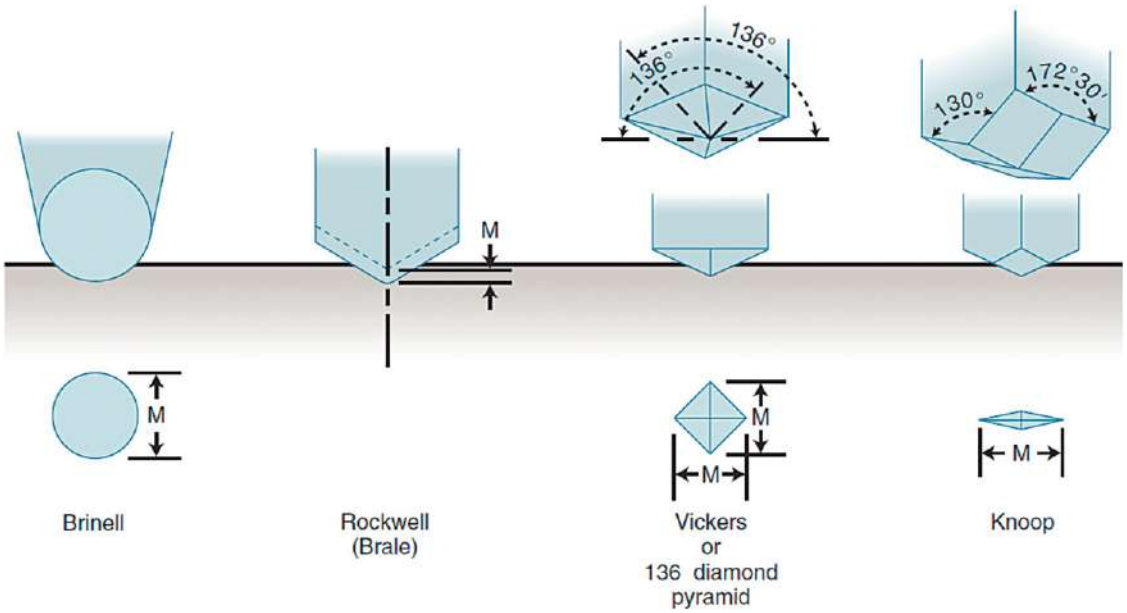


Figure 6.15 Shapes of hardness indenter points *(upper row)* and the indentation depressions left in material surfaces *(lower row)*. The measured dimension *M* that is shown for each test is used to calculate hardness. The following tests are shown: Brinell test—a steel ball is used, and the diameter of the indentation is measured after removal of the indenter. Rockwell test—a conical indenter is impressed into the surface under a minor load *(dashed line)* and a major load *(solid line),* and *M* is the difference between the two penetration depths. In the Vickers or 136-degree diamond pyramid test, a pyramidal point is used and the diagonal length of the indentation is measured. In the Knoop test, a rhombohedral pyramid diamond tip is used and the long axis of the indentation is measured.
(From Anusavice KJ, Shen C, Rawls RH: *Phillips' science of dental materials*, ed 12, St. Louis, MO, 2013, Elsevier.)

Hardness Testing

The surface hardness of a material is often of interest and is determined by using some form of indenter. Basically, the indenting tool is dropped onto the surface of the material with a specified load and the size of the defect created on the material is measured using a microscope. There are different hardness tests that are optimally used with specific materials. These include Knoop, Vickers, Barcol, Brinell and Rockwell hardness testing (Fig. 6.15). The difference between these testers is the shape and material of the indenter and the force used.

One area where hardness testing is frequently used is in determining the degree of polymerization of composite resin materials. *The harder the surface, the greater the degree of conversion*. Studies generally compare the hardness of the top surface (closest to the light) and the hardness of the bottom surface. Some of the variables that have been tested include intensity of the light, thickness of the composite resin increment, curing through ceramics and distance of the light from the resin.

Bond Strength Testing

In the adhesive era, there is a need to test the bond of one material to another, and the bond of materials to enamel and dentin. There have been hundreds of studies published on bond strength in recent years. These studies are useful for comparing the relative bond strength of product X to product Y, but have relatively low ability to predict clinical performance.[11]

The most common bond strength test is the *shear bond strength test*. This test is used frequently because it is easy to perform, but the clinical relevance is minimal. Another approach is the *microtensile bond strength test*. This test became popular because it was initially thought that 8–10 samples could be obtained from a single tooth; however, subsequent evaluation determined that data from these samples needed to be combined and in reality only 1 sample was obtained from each tooth. This bond strength test is extremely technique sensitive and produces data with wide standard deviations. It is not clear that the predictive data from microtensile bond strength testing are more predictive of clinical performance than shear bond testing.

It is clear that many bonds are quite strong initially, but deteriorate over time. Thus, it is imperative that bond strength testing must include initial testing plus testing after storage in water, thermocycling or fatigue.

Biologic Properties

Biologic properties of biomaterials are concerned with *toxicity* and *sensitivity* reactions that occur locally, within the associated tissue, and systemically. It is possible to evaluate

local toxic effects on cells by clinical pulp studies or by tissue culture tests. Unset materials may release cytotoxic components; however, in clinical situations this problem is rarely evident. Two important clinical factors determining toxicity are the *exposure time* and the *concentration* of the potentially toxic substance.

Generally, restorative materials harden quickly or are not readily soluble in tissue fluids (or both). Potentially, toxic products do not have time to diffuse into tissues. Even more importantly, the concentration makes the poison. Some authorities believe that if the amount of material involved is small, the pulp or other tissues can transport and excrete it without significant biochemical damage. Others believe that there is no threshold. A threshold level for toxicity is one below which no effect can be detected.

This section is not meant to be an exhaustive description of all physical and mechanical properties of dental materials. It is intended to highlight some of the more important properties and to describe some clinical examples where these properties might be important. Even though these properties are important, it is imperative to understand that in vitro studies of physical and mechanical properties are poor indicators of clinical performance. As fatigue studies become more sophisticated and clinically relevant, in vitro studies may become more predictive.

Clinical Notes

I. A large number of physical and mechanical properties of dental materials can be evaluated with in vitro testing.
II. When testing materials, whenever possible they should be tested in the mode in which they are used.
III. Bond strength studies have little relevance to clinical performance.
IV. Static in vitro studies of physical and mechanical properties yield minimal data that are predictive of clinical performance.
V. Ceramic materials should be fatigue tested under water.
VI. Fatigue testing with cyclic loading is promising but complex.

Direct Restorative Materials

Silver Amalgam

This section is discussed in detail in Chapter 12

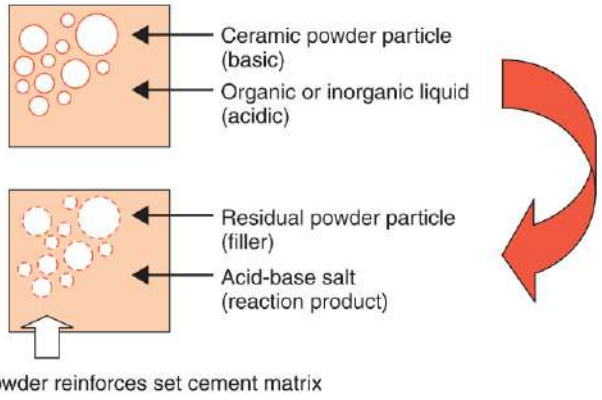


Figure 6.16 Schematic representation of dental cement components and microstructures. An acid-functional liquid is mixed with a basic powder. Reaction of the periphery of the powder consumes the acid groups of the liquid, producing a reaction product matrix that surrounds residual powder particles. The microstructure is a classic composite-like one with residual powder particles reinforcing the weaker matrix (Courtesy S C Bayne, School of Dentistry, University of Michigan, Ann Arbor, MI).

Composite Resins

This section is discussed in detail in Chapter 16

Dental Cements

Dental cements can be classified on the basis of the core constituent into *ceramic*- and *polymer*-based dental cements. Traditional dental cements are based on reactions between acidic liquids and basic powders to produce reaction product salts that form a solid matrix surrounding residual powder particles (Fig. 6.16). Microscopically, these cements are classic examples of filler and matrix microstructures. In all cements, the properties of interest are governed by the extent to which the matrix is minimized in the final material.

Traditional ceramic dental cements and polymer-based dental cements are listed in Table 6.3, along with identification of the major powder and liquid components and the reaction products.

Zinc oxide-eugenol (ZOE), reinforced ZOE, ZOE-EBA, silicate and zinc silicophosphate traditional ceramic cements are no longer routinely used in permanent cement restorations. Zinc phosphate and polycarboxylate were used until about 1990 but now have been extensively replaced by RMGI, compomer and composite cements.

Zinc Phosphate

Zinc phosphate cement originally was developed more than 100 years ago and was extremely popular during most of the 20th century. For that reason, it is often referred to

Table 6.3 Summary of dental cement classifications, abbreviations, reactants and reaction products

Classifications	Abbreviations	Liquid components	Powder components	Reaction product matrix
Traditional ceramic dental cements				
Unmodified ZOE	ZOE	Eugenol	ZnO	Crystalline zinc eugenolate
Resin-reinforced ZOE	R-ZOE	Eugenol	ZnO, polymer, resin	Crystalline zinc eugenolate
EBA-modified ZOE	ZOE-EBA	Eugenol, EBA	ZnO, Al_2O_3, polymer	Crystalline zinc eugenolate, crystalline zinc ethoxybenzoate
Zinc phosphate	ZP	H_3PO_4, H_2O	ZnO	Crystalline tertiary zinc phosphate
Silicate	SC	H_3PO_4, H_2O	F-Al-silicate glass	Amorphous silicophosphate
Zinc silicophosphate	ZSP	H_3PO_4, H_2O	F-Al-Silicate glass, ZnO	Amorphous silicophosphate, crystalline tertiary zinc phosphate
Polymer-based dental cements				
Polycarboxylate	PC	PAA, H_2O	ZnO	Amorphous zinc-polyacrylate gel
Conventional GI	GI	PAA, H_2O	F-Al-silicate glass	Amorphous aluminopolyacrylate gel
Resin-modified GI	RMGI	PAA, H_2O, water-soluble monomers	F-Al-silicate glass	Amorphous aluminopolyacrylate gel, cross-linked polymer
Compomer	CM	Monomers	F-Al-silicate glass	Amorphous cross-linked polymer, aluminopolyacrylate gel
Composite (or resin)	CP	Monomers	Silicate glass	Amorphous cross-linked polymer

EBA, ethoxybenzoic acid; *GI*, glass ionomer; *PAA*, polyacrylic acid; *ZOE*, zinc oxide–eugenol.

as the 'gold standard' for all dental cements despite the fact that its laboratory properties are generally inferior to those of most other currently used cements.

Composition

- **Powder:** 90% zinc oxide powder with 10% magnesium oxide
- **Liquid:** 50% phosphoric acid in water and is buffered with aluminium and zinc salts to control the pH

Manipulation

I. Components are mixed typically 2: 1 (powder-to-liquid by weight) on a chilled glass slab using controlled additions of powder to the liquid.
II. These precautions are necessary to reduce reaction speed, alter the pH in a controlled manner, dissipate heat from the exothermic reaction and provide sufficient working time.

Setting reaction

During setting, phosphoric acid ions react with zinc ions to produce a successive series of hydrated zinc phosphate salts. Ultimately, this results in tertiary zinc phosphate crystals that form a crystalline matrix around residual polycrystalline zinc oxide particles.

Zinc phosphate tends to be weaker and more brittle than polymer-based dental cements that have amorphous matrices.

Polycarboxylate Cement

Polycarboxylate cement was developed in the 1960s by Smith in an effort to:

I. Circumvent potential pulpal problems associated with the low pH of traditional cements (e.g. zinc phosphate cement) and biocompatibility problems related to the mobility of small acidic ions.
II. By choosing an acid-functional polymer as a substitute for phosphoric acid in forming the matrix, it was also possible to produce cements that could *adhere via chelation* to dental surfaces.

The original acid-functional polymer was *polyacrylic acid*. More recent commercial products include two or more monomers in the polymer. It is technically more correct to refer to the final polymer as a *polyalkenoate*. Cements based on solutions of these polymers (i.e. polycarboxylate cement or GIC) may be called *polyalkenoic cements*.

Glass Ionomer Cements

GIC are hybrids of silicate and polycarboxylate cements designed to combine the optical and fluoride-releasing properties of silicate particles with the chemically adhesive and more biocompatible characteristics of the polyacrylic acid matrix compared with the extremely acidic matrix of silicate cement. Glass ionomers (GIs) are explained in detail later in this chapter.

Resin-Modified Glass Ionomer (RMGI) Cements

Resin-Modified Glass Ionomer (RMGI) cements have the liquid and powder components modified to provide visible light–curing reactions in addition to traditional GI setting reactions. RMGIs have been extremely popular since the 1990s.

Composite Resin Cements

Composite cements (or resin cements or composite resin cements) have the same general components as composite restorative materials but generally have slightly lower concentrations of filler particles.

These materials are reserved almost exclusively for use with all-ceramic restorations, although they are suitable for use with all indirect restorative materials. These materials have the *best laboratory properties of all cements* but require more complicated clinical procedures and generally include bonding systems for dentin, enamel and the restoration.

Some newer resin cements have been described as 'self-adhesive'. These cements have components similar to those used in self-etching dentin primers and thus have the potential to bond directly to clean dentin without separate binding procedures.

Clinical Applications of Dental Cements

I. Dental cements are most often used for *luting* indirect restorations. As luting agents, the most important clinical requirements are *flow*, *wetting* and *film thickness*.
   i. To enhance flow, the materials are mixed at relatively low powder-to-liquid ratio.
   ii. To guarantee that a thin film can be produced, dispersed phase particles of 5 µm or less in diameter should be used. The actual film thickness that is achieved ranges from *50–100* µm and depends on:
      – Viscosity of the mixture
      – Availability of space for displacement of the cement

II. Dental cements are also used as *bases*.

Although low powder-to-liquid ratio produce low viscosities for luting agents, cements used for bases should be mechanically stronger and are mixed with the maximum powder content manageable.

Composition, Structure and Properties

The final properties of dental cements depend on the powder-to-liquid ratio used during mixing. Higher powder-to-liquid ratio not only increases the mechanical strength but also increases the viscosity and reduces wetting and flow. The final matrices of the set cement are indicated in Table 6.3. A brief summary of dental cement properties is provided in Table 6.4.

Table 6.4 Characteristic properties of categories of luting dental cements*

	R-ZOE	EBA	ZP	PC	GI	RMGI	COMP
Working characteristics							
P/L ratio	[Low]	[Low]	[Low]	[Low]	[Low]	1.2–1.6	—
Film thickness (µm, ADA flow test)	32	—	18	21	24	—	10–60
Setting-time range (min)	6–8	—	5–7	2–3	3–5	2–3	—
Physical properties							
LCTE (ppm/°C)	[Low]	[Low]	[Low]	—	—	[Low]	50
Thermal conductivity	[Low]	[Low]	[Low]	[Low]	[Low]	[Low]	[Low]
Chemical properties							
Solubility/disintegration (%, ADA test)	0.08	0.05	0.06	0.60	1.25	—	0.01
Mechanical properties							
Compressive strength (MPa)	48	65	160	70	120	148–180	170–190
Diametral tensile strength (MPa)	4	7	10	10	12	30–35	30–35
Biological properties							
Pulpal response	[Mild]	[Mild]	[Moderate]	[Moderate]	[Mild]	[Mild]	—

ADA, American Dental Association; *LCTE*, linear coefficient of thermal expansion; *MPa*, megapascal; *P/L*, powder-to-liquid. (Table 22.1, for abbreviations for cement).
*Relative values are shown in brackets. These values are representative of a wide range of possible values.

Clinical Notes

- Zinc phosphate cements have the potential during setting to release components from the acid-rich matrix into dentin and irritate the pulp. When using zinc phosphate cement, dentin is routinely protected with a cavity sealer.
- Other cements that can chemically bond to dentin must, however, be allowed to come into direct contact with dentin. Cavity sealers should not be used to coat dentin if a polycarboxylate or conventional GIC is to be used for chemical adhesion.
- When bonding composite or ceramic restorations, dentin should not be sealed with cavity sealers, but instead the entire preparation should be treated with a bonding system.

Objectives for Pulpal Protection

Many restorative biomaterials that provide excellent properties for the bulk of a dental restoration may not protect the dental pulp during setting or during cyclic, thermal or mechanical stressing. Pulpal protection (Fig. 6.17) requires consideration of *sealing the smear layer, chemical, electrical, thermal and mechanical protection as well as pulpal medication.*

Sealing the smear layer

Tooth preparation with rotary instruments generates cutting debris, some of which is compacted unavoidably into a layer on the cut surface. This layer of material is called *smear layer* and is typical of any cut surface, dental or otherwise. The dentinal smear layer (Fig. 6.18) produces some degree of dentinal tubule sealing. However, it is 25%–30% porous and cannot prevent slow long-term diffusion. For amalgam restorations that can leak along their enamel margins, the smear layer should be sealed to produce chemical protection.

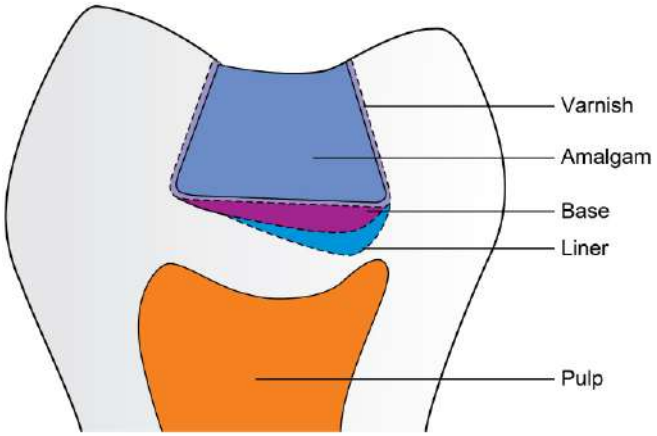


Figure 6.17 Schematic view of needs for pulpal protection below the metallic restoration. Varnishes, liners and bases may be added to the tooth preparations under amalgam for purposes of chemical, electrical, thermal or mechanical protection or pulpal medication.

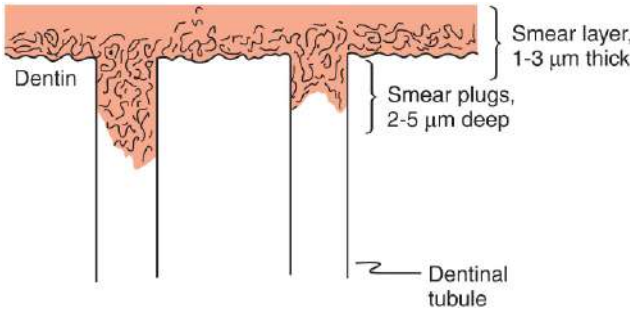


Figure 6.18 Schematic view of the dentin smear layer.

Thermal insulation

A key function of enamel and dentin is thermal insulation of the pulp. Most restorative materials are not as insulating as dentin, and thermal insults may occur during intraoral temperature changes. The need for insulation is greatest for metallic restorations. Thermal insulation is proportional to the thickness of the insulating material. Approximately, *2 mm of dentin, or an equivalent thickness of material*, should exist to protect the pulp (Fig. 6.19). This thickness is not always possible, but 1–1.5 mm of insulation is accepted as a practical thickness. As the tooth preparation extends closer to the pulp, a thick liner or a base is used to augment dentin to the proper thickness range. Such a liner or base cannot harden by evaporation of solvent or water because it would not dry effectively. Material used for this purpose hardens by a chemical reaction or is light-cured.

Pulpal medication

In addition to thermal protection, liners are formulated to provide pulpal medication whenever possible. Two important aspects of pulpal medication are the *relief of pulpal inflammation* and *facilitation of dentinal bridging* for physiologic protection. The materials (calcium hydroxide and eugenol) most commonly used to provide these two functions are not mutually compatible and cannot be used in the same formulation.

Mechanical protection

As the depth of the tooth preparation increases, the amount of remaining dentin thickness reduces. Hence, in deep preparations a thicker base that provides supplemental mechanical protection is recommended. This mechanical support provides resistance against disruption

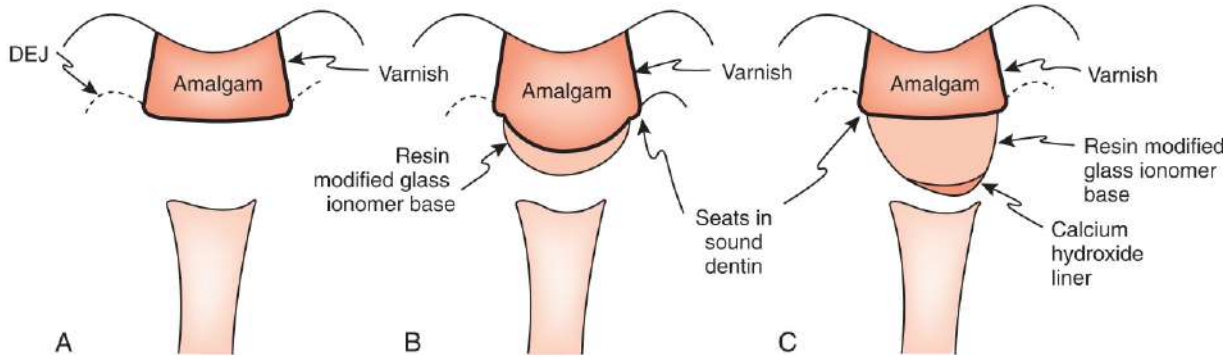


Figure 6.19 Schematic examples of use of liners and bases for amalgam restorations. (A) For shallow amalgam tooth preparations, varnish or sealer is applied to walls of preparation before insertion of restoration. (B) For moderate-depth tooth preparations, light cured RMGI may be placed for thermal protection. (Note the seats in sound dentin for amalgam restoration.) (C) In very deep preparations, light-cured calcium hydroxide is placed in the deepest region in which infected dentin was excavated, and then the base of RMGI is inserted.

of thin dentin over the pulp during amalgam condensation procedures or cementation procedures of indirect restorations.

Bases and Liners

Bases and cavity liners have been used for many years in restorative dentistry. With the development of more effective adhesives, the use of materials as bases and liners has diminished. The topic of bases and cavity liners is quite controversial today and there is a general lack of consensus among authorities, practitioners and educational institutions as to indications for such materials.[12] The following is a brief description of materials used as bases, liners, varnishes and sealing agents available today.

Liners

Liners are relatively thin layers of material used primarily to provide a barrier to protect dentin from residual reactants diffusing out of a restoration or from oral fluids (or both) that may penetrate leaky tooth restoration interfaces. They also contribute *initial electrical insulation*, generate *some thermal protection* and, in some formulations, provide pulpal treatment (Fig. 6.17).

Liners can be classified on the basis of their film thickness into:

I. Thin film liners (2–5 μm), which can be subdivided into:
   i. Cavity varnish
   ii. Cavity sealer
II. Cavity liners (thicker liners of 200–500 μm = 0.2–0.5 mm).

Thin Film Liners:

I. **Cavity varnish**: Cavity varnish is a solution of a natural gum or synthetic resin in an organic solvent. Copal varnish (Copalite, Temrex, Freeport, NY) has been traditionally used under amalgam restorations to reduce initial levels of marginal leakage. *Copalite* contains 10% copal resin in a combination of ether, alcohol and acetone. ***Cavity varnishes are no longer popular and have been replaced by cavity sealers.***
II. **Cavity sealer**: A cavity sealer is a liquid primer-like material that seals the dentin surface prior to placing definitive restorative materials. Current products used as cavity sealers under silver amalgam restorations are *Gluma Desensitizer* (Kulzer, South Bend, IN) and *G5 desensitizer* (Clinician's Choice, New Milford, CT). These sealers contain both HEMA and Glutaraldehyde and are very effective sealing agents for dentin.

Cavity Liner

A **cavity liner** (***200–500 μm***) is a fluid paste applied in a thin layer as a protective barrier between dentin and the restorative material. Liners should *not* be used in layers thicker than 0.5 mm.

The two most commonly used cavity liners are:

I. *Calcium hydroxide*:
   Calcium-hydroxide liners (e.g. Dycal, Dentsply Sirona, York, PA) have a very high pH (9–14), are antibacterial and promote sclerosis, hence reduce permeability. They must not be used as a base because they have very poor physical properties and should only be placed as a very thin (0.5 mm) liner over the deepest portion of the cavity preparation. Calcium hydroxide materials have traditionally been used when performing a direct pulp cap, as the high pH irritates the pulp tissues and stimulates the production of secondary dentin. Calcium hydroxide has largely been replaced by bioactive cements such as mineral trioxide aggregate (MTA) and biosilicate cement (Biodentine, Lancaster, PA) for this function.
   If used, it is recommended that a calcium-hydroxide liner be overlaid with an RMGI base.
II. *RMGI*:
   RMGI liners (e.g. Vitrebond and Vitrebond Plus, 3M Oral Care, St. Paul, MN; Fuji Lining Cement, GC

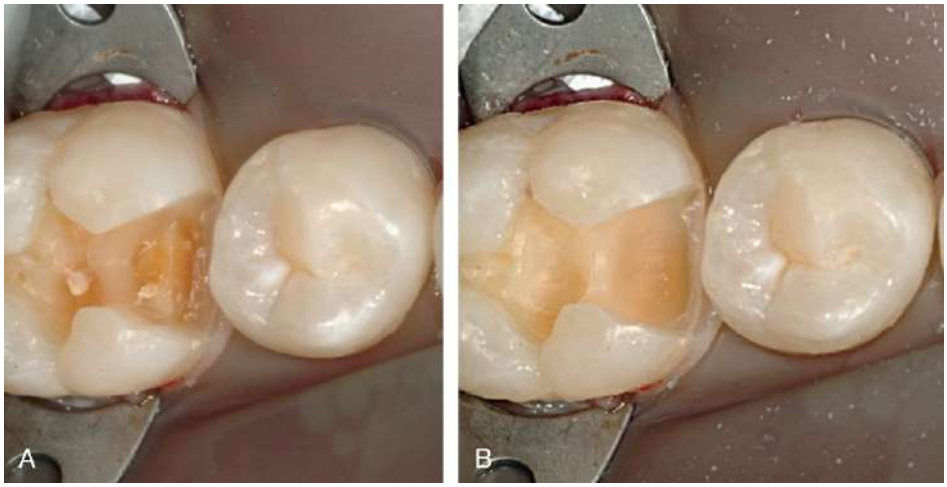


Figure 6.20 (A) Cavity preparation of tooth 30 before placing cavity liner. (B) Preparation after placement of cavity liner.

America, Alsip, IL) are supplied as powder/liquid or paste/paste systems. They are light activated and have improved handling characteristics and physical properties compared to regular GI lining materials. They bond very predictably to dentin, provide an excellent seal and are very compatible with the pulp. Typically, they are used as a thin liner (0.5 mm) over dentin in deep cavity preparations (Figs. 6.19 and 6.20).

The primary purpose of the liners is to provide a protective seal on the exposed dentinal surface. The liner layer at the restoration–enamel interface also provides a means of electrically isolating metallic restorations from external electrical circuits with restorations in the adjacent teeth.

Clinical Notes

- The need for liners is greatest with pulpally extended metallic restorations that are not well bonded to the tooth structure and that are not insulating, such as amalgam and cast gold, or with other indirect restorations.
- Direct composite restorations, indirect composite or ceramic restorations, and RMGI restorations routinely are bonded to the tooth structure. The insulating nature of these tooth-coloured materials and the sealing effects of the bonding agents preclude the need for traditional liners and bases, unless the tooth preparation is extremely close to the pulp, and pulpal medication becomes a concern.

Bases

A **base** is any substance (typically 1–2 mm) placed under a restoration that blocks out undercuts in the preparation, acts as a thermal or chemical barrier to the pulp and/or controls the thickness of the overlying restoration (Fig. 6.21).



Figure 6.21 Base material used to block out undercuts in the preparation of maxillary anterior teeth.

Bases are used to provide *thermal protection* for the pulp and to *supplement mechanical support* for the restoration by distributing local stresses from the restoration across the underlying dentinal surface.

Any dental cement mixed to a very thick consistency can be used as a base. The primary contemporary function of a base is to block out undercuts for indirect restorations, to seal the dentin against leakage in deep cavities, and to act as replacement dentin with large composite resin restorations to reduce the volume of composite resin.

Types of bases

I. Highly modified forms of GIC (*light-cured RMGIs or compomers*) provide chemical adhesion, good mechanical strength, potential fluoride release, well-controlled setting and rapid achievement of strength.
II. *GIC* became more popular from 1985 to 1994.
III. *Polycarboxylate cement* bases gained popularity starting in the 1970s.
IV. *Zinc phosphate cement* and *resin-reinforced ZOE cement* were widely used for bases before the 1960s.

Clinical Notes

- Metallic restorations may benefit from seating (resting) on sound dentin peripheral to the lined or based regions that result from excavating infected dentin (Fig. 6.22). These seats may help distribute stresses laterally to sound dentin and away from weaker underlying structures.
- Various liners and bases may be combined in a single preparation, and the dimension between the restoration and the pulp may be a combination of natural dentin, liner and base.
- Newer liners place less emphasis on pulpal medication and focus more on chemical protection by sealing, adhesion and mechanical protection. *Sealing may prove to be the most important property overall.*
- Before the development of RMGIs, the functions of liners and bases were relatively distinct but have since begun to converge.

Composition, Structure and Properties

Representative examples of the composition, structure and important properties of solution cavity varnishes, cavity liners and bases are presented in Tables 6.5,6.6 and 6.7.

Table 6.5 Composition, structure and properties of a typical solution liner (varnish)*

	Copal resin varnish (Copalite)
Components	
Solid (10%)	Copal resin
Solvent (90%)	Ether, acetone, alcohol
Setting reaction	Physical (evaporation)
Structure	
Arrangement	Amorphous film
Bonding	Covalently bonded
Composition (phases)	Single phase
Defects	Pores and cracks
Physical properties	
Thermal	[Insulator]
Electrical	[Insulator]
LCTE (ppm/°C)	[High]
Wetting	[Poor]
Chemical properties	
Solubility (% in water)	[Low]
Mechanical properties	
Tensile strength (MPa)	<1
Elongation (%)	<0.1%
Biologic properties	
Toxicity	[None, if solvent eliminated safely]

LCTE, Linear coefficient of thermal expansion.
*Relative properties are reported in brackets.

Clinical Considerations

I. Based on Remaining Dentin Thickness

Clinical judgments about the need for specific liners and bases are linked to the amount of *remaining dentin thickness* (*RDT*), considerations of adhesive materials and the type of restorative material being used. Recommendations for various restorative procedures are summarized in Table 6.8.

I. **Shallow preparation (≥2 mm of RDT)**

In a shallow tooth excavation (which includes ≥2 mm of RDT), pulpal protection, other than in terms of chemical protection, is not necessary.

i. For an amalgam restoration, the preparation is coated with *two thin coats of a varnish, a single coat of a dentin sealer or a dentin bonding system*, and then restored. In most cases, a *dentin sealer* is the material of choice (e.g. Gluma by Heraeus Kulzer, South Bend, IN; Hurriseal by Beutlich Pharmaceuticals, Waukegan, IL).

ii. For a composite restoration, the preparation is treated with a *bonding system* (etched, primed, coated bonding agent) and then restored.

II. **Moderately deep preparation (0.5–2 mm of RDT)**

i. In a moderately deep tooth excavation for amalgam that includes some extension of the preparation toward the pulp so that a region includes less than ideal dentin protection, it may be judicious to apply a base *only at that site* using light cured RMGI.

ii. Traditionally, zinc phosphate cement and polycarboxylate cement were the preferred bases under amalgam restorations. However, *light cured RMGIs are now recommended*.

iii. In a composite tooth preparation, eugenol has the potential to inhibit polymerization of layers of bonding agent or composite in contact with it. RMGI is normally used if a base is indicated.

III. **Very deep preparation (<0.5 mm of RDT)**

i. If the RDT is very small or if pulp exposure is a potential problem, calcium hydroxide is used to stimulate reparative dentin for any restorative material. A thickness of 0.5–1 mm of set calcium hydroxide liner is sufficient to treat a near or actual pulp exposure.

ii. If extensive dentin is lost because of caries, and the tooth excavation extends close to the pulp, a *light cured RMGI base should be applied over the already placed calcium hydroxide liner*.

iii. The sealer or bonding agent is not applied until after the base is in place.

Table 6.6 Composition, structure and properties of typical liners*

	Calcium hydroxide (VLC Dycal)	Traditional GI (Fuji Lining LC)	Reinforced ZOE (IRM)
Components			
Components 1 and 2	Paste (with $Ca(OH)_2$, LC resin, and polyphenolics)	Powder (Al-silicate glass); liquid (polyalkenoate acid, LC resin)	Paste (with ZnO); paste (with eugenol)
P/L or paste/paste ratio	(1 component)	1.4/1 by weight	6/1 by weight
Setting reaction	Acid-base reaction	Acid-base reaction	Acid-base reaction
Structure			
Arrangement	Amorphous matrix Crystalline fillers	Amorphous matrix Crystalline fillers	Crystalline matrix Crystalline fillers
Bonding	Covalent; ionic	Covalent; ionic	Covalent; ionic
Composition (phases)	Multiphase	Multiphase	Multiphase
Defects	Pores; cracks	Pores; cracks	Pores; cracks
Physical properties			
LCTE (ppm/°C)	[Low]	[Low]	[Low]
Thermal conductivity	[Insulator]	[Insulator]	[Insulator]
Electrical conductivity	[Insulator]	[Insulator]	[Insulator]
Radiopacity (mm Al)	—	4	—
Chemical properties			
Solubility (% in water)	0.3–0.5 [high]	0.08 [low]	[Modest]
Shrinkage on setting (μm/mm)	—	24 [low]	—
Mechanical properties			
Elastic modulus (MPa)	588	1820	—
Hardness (KHN_{100})	—	—	—
Elongation (%)	—	—	—
Compressive strength, >24 h (MPa)	138	128	71
Diametral tensile strength (MPa)	—	24	—
Flexural strength (MPa)	—	46	—
Dentin shear bond (MPa)	—	5.8	—
Biologic properties			
Biocompatibility	[Acceptable]	[Acceptable]	[Acceptable]

LCTE, linear coefficient of thermal expansion; *MPa*, megapascal; *ppm*, parts per million; *ZOE*, zinc oxide-eugenol.
*Relative properties are shown in brackets. The values reported are from a variety of published sources from 1988 to 2000, including manufacturer's product bulletins. Comparisons should be made only in terms of the overall application requirements and not in terms of any single property

II. Based on kind of Final Restoration

When using bases and liners, it should be remembered that the ultimate goal is to have a minimum of 2 mm of tooth structure or a combination of tooth structure base and liner between the restorative material and the pulp.[13]

I. **Amalgam**
   i. It is no longer considered necessary to use bases to build up deep cavity preparations to 'ideal' form. With deep preparations close to the pulp, a very small liner of calcium hydroxide may be placed, followed by a thin liner of RMGI (Vitrebond Plus). As an alternative, the calcium hydroxide liner may be omitted. Deep preparations that are not close to the pulp should receive a thin RMGI liner.
   ii. Gluma Desensitizer or G5 should be applied to all preparations, including those in which liners have been placed.

II. **Composite Resin**
   i. For any preparation deeper than a minimal preparation, a thin RMGI liner should be placed in the deepest portion of the preparation. In this situation (deep dentin), dentin adhesives tend to be less effective because there is less intertubular dentin to interact with and form an effective hybrid layer.

Table 6.7 Composition, structure and properties of typical bases*

	Resin-modified glass ionomer (RMGI) cement (Vitremer)	Zinc phosphate cement (modern Tenacin)	Polycarboxylate cement (Durelon)	Glass ionomer cement (Ketac-Cem)
Components				
Component 1	F-Al-Si glass powder†	ZnO powder	ZnO powder	F-Al-Si glass powder
Component 2	Monomers‡/H_2O	H_3PO_4/H_2O	Polyacrylic acid/H_2O	Polyacrylic acid/H_2O
P/L ratio	2.5	[High]	[High]	[High]
Setting reaction	Acid-base; free radical	Acid-base	Acid-base	Acid-base
Structure				
Arrangement	Amorphous matrix	Crystalline matrix	Amorphous matrix	Amorphous matrix
	Glass fillers	Crystalline fillers	Glass fillers	Crystalline fillers
Bonding	Covalent; ionic	Ionic	Covalent; ionic	Covalent; ionic
Composition (phases)	Multiphase	Multiphase	Multiphase	Multiphase
Defects	Pores and cracks	Pores and cracks	Pores and cracks	Pores and cracks
Physical properties				
Thermal	[Insulator]	[Insulator]	[Insulator]	[Insulator]
Electrical	[Insulator]	[Insulator]	[Insulator]	[Insulator]
LCTE (ppm/°C)	[Low]	[Low]	[Low]	10 [low]
Chemical properties				
Solubility (% in water)	0.2 [low]	0.10 [low]	[Low]	0.70 [low]
Mechanical properties				
Modulus (MPa)	—	—	—	—
Hardness (KHN_{100})	—	—	—	—
Percent elongation (%)	—	—	—	—
Compressive strength (MPa)	200	77	[100]	120
Diametral tensile strength (MPa)	35	—	[17]	—
Biologic properties				
Safety	[Acceptable]	[Acceptable]	[Acceptable]	[Acceptable]

*Relative or estimated properties are shown in brackets.
†Including a redox catalyst.
‡Polycarboxylic acid/HEMA/methacrylates/water/ethanol/photo-initiator.

Table 6.8 Summary of pulpal protection procedures (medicament/liner/sealer)

	Shallow excavation (RDT > 2 mm)	Moderate excavation (RDT 0.5–2 mm)	Deep excavation (RDT < 0.5 mm)
Amalgam	No/No/sealer	No/base/sealer	CH/base/sealer
Composite	No/No/DBS	No/No/DBS	CH/No/DBS
Gold inlays and onlays	No/No/cement	No/base/cement	CH/base/cement
Ceramic, PR, FRP	No/No/DBS, CC	No/No/DBS, CC	CH/No/DBS, CC

Note: Pulpal protection includes pulpal medication, dentin sealing, thermal insulation, electrical insulation, and mechanical protection. Sealer = Gluma, Hurriseal, or others; base = Vitrebond, Durelon, or others; cement resin-modified glass ionomer. *CC*, Composite cement (e.g. Rely X Luting Cement); *CH*, Dycal liner; *DBS*, dentin-bonding system; *FRP*, fiber-reinforced prosthesis; *PR*, processed resin; *RDT*, remaining dentin thickness.

ii. With posterior composite resin preparations where the proximal box ends close to or beyond the CEJ, a RMGI liner should be placed to reduce postoperative sensitivity.
iii. Adhesive bonding systems *must* be used with all composite resin restorations regardless of whether a lining material has been placed.

III. **Cast Metal Indirect Restorations**
i. Deep restorations should receive a RMGI liner prior to impression making. With very deep lesions, vitality testing must be done before administration of local anesthetic to determine the need for endodontic therapy. RMGI, Fuji IX or other base materials can be used to block out undercuts.
ii. Gluma Desensitizer or G5 should be applied immediately after tooth preparation prior to making the provisional restoration.

Clinical Notes

I. There is still a need for bases and liners in the adhesive era.
II. Bases are primarily used to block out undercuts.
III. Liners should be used in a thin film, no greater than 0.5 mm thick.
IV. RMGIs are used with deep caries prior to bonding because bonding to deep dentin with available dentin bonding agents is not predictable due to the lack of intertubular dentin.
V. Cavity sealers should be used routinely under amalgam restorations.

Glass Ionomer Restorative Materials

GIs were introduced to the profession by Wilson and Kent in the early 1970s, and since then have evolved into a number of unique dental materials.[14] This section describes available GI products and also provides specific indications and contraindications for the clinical use of GI materials (Fig. 6.22).



Figure 6.22 Glass ionomer restorative material. (Courtesy GC.)

Box 6.1 Setting reaction of glass ionomer cement

The acidic liquid solution (pH = 1) dissolves portions of the periphery of the silicate glass particle, releasing calcium, aluminum, fluoride, silicon and other ions
↓
Calcium ions are chelated quickly by carboxyl side groups on polyacrylic acid polymer chains
↓
Cross-linking of the polyacrylic acid chains takes place producing an amorphous polymer gel
↓
During the next 24–72 h, the calcium ions are replaced by more slowly reacting aluminium ions to produce a more highly cross-linked matrix that is now mechanically stronger
↓
It is now believed that during the maturation involving aluminium ion cross-linking, silicon ions and unbound water participate in producing an inorganic co-matrix, best described as a *hydrated silicate*
↓
The same carboxylic acid side groups also are capable of chelating surface ions on the calcium ions from the tooth structure
↓
This process generates true chemical bonds at all internal and external interfaces when the reaction conditions are correct

GI materials are indicated to be employed as:

I. Dental cements/luting agents
II. Restorative materials (primarily Class V)
III. Cavity liners
IV. Bases

Based on clinical trials and physical properties testing, the efficacy of GI for pit-and-fissure sealants and for large core build-ups is questionable.

All GI materials have a number of features in common. While some materials have multiple setting mechanisms, the basic GI setting reaction is an acid-base reaction with polyacrylic acid acting on and partially dissolving aluminosilicate glass (Box 6.1).

This results in a 'burst' of fluoride release at a very high level, but this burst of fluoride release is completed in 24 h. After that, there is a continual low-level release of fluoride that is likely well below the threshold necessary to protect against secondary caries. Most publications on GI materials state that GI is a fluoride releasing material, which implies protection from secondary caries. However, GI materials must be 'recharged' with fluoride, which can then be re-released in order to provide protection. This ability for GI materials to act as 'reservoirs' for fluoride will be discussed later.

GI materials form very predictable bonds to both enamel and dentin. The bonds are weak chemical (chelation) bonds, but are predictable, effective and durable

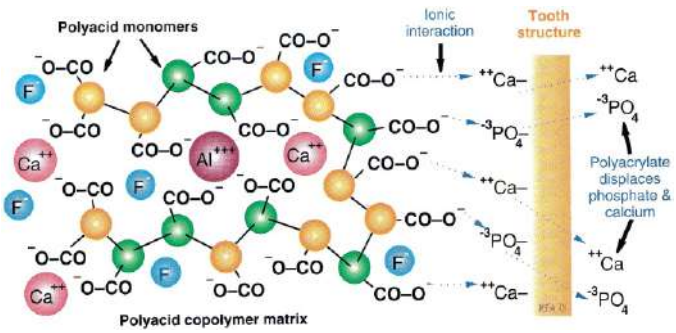


Figure 6.23 Schematic illustration of ionic interaction between glass ionomer ions and tooth structure.
(From Albers HF: *Tooth-colored restoratives: principles and techniques*, 2002, PMPH-USA.)

(Fig. 6.23). In general, GI materials have inferior physical properties compared to amalgam and resin composite.

Types of glass ionomer materials

I. Conventional glass ionomer cements

- *Composition:*
  - **Powder:** Complex *aluminosilicate powder* containing calcium and fluoride.
  - **Liquid:** Solution of polyacrylic acid liquid modified by co-polymerization with small amounts of *itaconic acid* and *tartaric acid* to increase the stability of the liquid and modify its reactivity.

  Examples include Fuji Type I and Ketac-Cem. The setting reaction is a chemical acid-base reaction.
- *Clinical Considerations:*

  These cements bond to tooth structure and have good physical properties compared to the 'gold standard' cement, zinc phosphate. They release fluoride, although it is doubtful that this is of any clinical significance.

  The main disadvantage of these cements is that they have been associated anecdotally with an increased incidence of postoperative sensitivity. Clinical trials have demonstrated that this sensitivity is likely related to technique (e.g. overdrying the prepared tooth) or contamination of the setting cement by saliva. With proper technique, the level of post-cementation sensitivity should be very low.

Conventional GI cements have decreased in popularity in recent years because of the development of resin-modified GI cements.

II. Resin-Modified Glass Ionomer Cements

Composition and setting reaction:

RMGI cements are the most popular cements available today. Examples include Rely-X Luting and Rely-X Luting Plus cements, Fuji Plus and Fuji-Cem cements. Resin monomers have been incorporated into these materials, and thus these cements have two setting reactions: the *classic acid-base reaction* and a *chemical-cure resin reaction*.

Clinical Considerations:

Resin-modified GI cements bond to tooth structure and possess the same fluoride release characteristics as conventional GI cements. As a result of the addition of resin, they possess:

I. Superior physical properties
II. Improved handling characteristics
III. Less technique sensitive

Examples of a traditional GI and RMGI, are listed in Table 6.9 along with certain key properties.

Resin-modified GI cements have almost totally replaced zinc phosphate cement and are indicated for cementation of gold inlays, onlays, partial veneer and full veneer crowns and fixed partial dentures, metal-ceramic crowns and fixed partial dentures and monolithic and layered zirconia crowns and fixed partial dentures. The manipulation of RMGI cements is relatively simple, making them relatively free of technique sensitivity. They are supplied with three different dispensing techniques (Fig. 6.24).

I. The original cement was supplied as a *powder and liquid*. It is important with powder/liquid cements to closely follow the manufacturer's directions regarding the ratio of powder to liquid.
II. A later version of RMGI was the development of a *clicker* dispenser. With this product, it was simply a matter of dispensing the correct amount of cement and mixing. This simplified approach was partially responsible for RMGI cements gaining the lion's share of the cement market.
III. The latest innovation has been the development of *automix* dispensers. With these devices the operator simply dispenses the mixed cement into the restoration and proceeds with cementation.

The prepared tooth should be cleaned with pumice or an intraoral air particle abrasion device if available. The use of a desensitizer such as Gluma Desensitizer or G5 is optional, but has been shown to not only *not* interfere with bonding, but actually slightly enhances it with RMGI cements. A *small* amount of cement should be placed in the restoration and the restoration should be seated with firm finger pressure and then completely seated. The cement should be protected from saliva contamination while setting and *should be allowed to set completely* before removing the excess cement.

Many manufacturers of RMGI cements and some brands of resin cements are suggesting that the operator should 'preset' the excess setting cement with 5 s exposure to a curing light to plasticize the excess cement. The excess cement can then be removed readily with one or two deft moves with an explorer. This is not recommended, as it is likely that cement just under the margin will also be removed along with the excess cement.

Table 6.9 Comparison of compositions, structures and properties of typical examples of conventional GI and RMGI cements

		Conventional glass ionomer (GI)	Resin-modified glass ionomer (RMGI)
Abbreviation		GI	RMGI
Commercial name		Fuji II	Vitremer
Manufacturer		GC	3M ESPE
Applications		Liner, base, cement	Cement, restorative
Acid-base setting reaction		Yes	Yes
Polymerization setting reaction		No	Yes
Properties			
VLC depth of cure (mm)		NA	2.7
Water absorption ($\mu g/mm^3$)	7 days	236	—
	180 days	—	174
Radiopacity (mm of Al)		2.5	1.8
Fluoride release ($\mu g/cm^2$)	7 days	25.9	21.2
	22 days	9.3	8.8
Flexural modulus (GPa)	Dry	12.9	9.6
	Wet	5.5	—
3-pt. flexure strength (MPa)	Dry	20	68
	Wet	4	—

III. Reinforced glass ionomers

A number of reinforced GI materials have been introduced over the years in an attempt to expand the indications for GI materials. The following is a partial listing of such materials:

I. *Miracle Mix* (Fuji II plus amalgam alloy particles)
   silver–tin particles (amalgam alloy particles) were admixed in some formulations to produce an amalgam substitute. This combination became known as the miracle mixture because it was initially introduced during the early 1980s at the time when the mercury controversy was increasing dentists' questions about the safety of amalgams.[15]
   The properties of the miracle mixture were far inferior to those of amalgam, however, and it was therefore not well received as a restorative material. In part, the problem with the admixture was that the matrix would not adhere strongly to the silver–tin alloy particles.



Figure 6.24 Three different dispensing methods for RMGI cements: powder liquid, clicker and automix.
(Courtesy ESPE.)

II. *Cermet Mixture*—Ketac Silver (a glass cermet where the silver particles are chemically attached to the glass)
   To circumvent the aforementioned difficulty, the silver–tin alloy particles were substituted by silver–palladium (Ag-Pd), which generates a passivating oxide film of palladium oxide that is chemically reactive by chelation with polyacrylic acid.
   These mixtures, termed as ceramic–metal (cermet) mixtures, were much stronger than unmodified GIC but had poor esthetics.

III. *Fuji IX and Ketac Molar* (self-cure GI with increased aluminosilicate glass concentrations, polyacrylic acid and polycarboxylic acid)

IV. *Equia and Equia Forte*

Manufacturers claim that these reinforced glass ionomers materials are indicated as core build-up materials for extensively damaged or endodontically treated teeth. The physical properties of these materials are ***inferior*** to silver amalgam and resin composite, and thus they should *not* be used for core build-ups. They can be used as bases and block-out materials.

Fuji IX is one of the most popular of these materials. It is indicated as a base under large composite resin restorations, as the intermediary material is an indirect pulp cap procedure, and as a temporary restorative material to replace missing restorations, broken cusps and so on. Fuji IX is mixed in an automix device. A conditioner of 10% polyacrylic acid is placed on the tooth for 20 s and washed off prior to placing Fuji IX.

IV. Polyacid-Modified Composites (Compomers)

Polyacid-modified composites or compomers were released in the 1990s. They are modestly filled composite resin materials to which has been added some pre-reacted GI material. Although the term *compomer* implies that the material possesses a combination of the characteristics of composites and GIs, these materials are essentially polymer-based composites that have been slightly modified to permit *fluoride release* from the glass or special matrix phases. They do release fluoride, but at a very low level, less than 10% of the release of GIs. Thus, it is unlikely that compomers have any meaningful effect on the incidence of secondary caries.

The important distinction between a GI product and a compomer is that there is ***no acid-base GI reaction*** with compomers. Examples of compomer products are Dyract, Compoglass and F2000.

Compomers are intended to be used in non-stress-bearing situations; and they have adequate esthetics, ease of handling, adhesion to tooth structure, *but poor physical properties*. There are likely no indications for use of compomers in contemporary dentistry.

V. ART Glass Ionomer

- An atraumatic restorative treatment (ART) restoration is based on a self-cured version of a conventional glass ionomer that is a mixture of powder and liquid and is capable of a relatively *high fluoride release*.
- Since 1990, a special filling material has been fabricated from GIs for use in the ART technique. In locations such as the rural regions of developing countries, where routine treatment is not possible, untrained dental personnel can use ART to halt or lessen the progression of frank carious lesions until the patient can access dental facilities.
- This technique and material is not recommended in routine clinical use and is restricted to geographical areas with limited access to dental treatment.

VI. Giomer

- In an attempt to retain some traditional properties of GIs, giomers include *precured* and *pulverized* particles of GI as an additional dispersed phase within a compomer.
- Early clinical trials seem to indicate that they are not truly competitive with composites as permanent filling materials in posterior locations.

Clinical Applications of Glass Ionomers

Restorative Materials

GI materials have been formulated for use as restorative materials, with their primary function being:

I. Restoration of noncarious cervical lesions
II. Root caries lesions

The original materials were chemically similar to GI cements with higher glass content and increased particle size. Examples of these materials would be Fuji II and Ketac Cem. These materials have essentially been replaced with RMGI restorative materials, which are less technique sensitive, have better physical properties, show improved esthetics and display vastly improved handling characteristics. Examples of RMGI restorative materials are Fuji II LC, Fuji Filling LC, Vitremer restorative material and Ketac Nano. The setting reactions of these materials are complex. All undergo the classic GI acid-base reaction as well as a light-activated resin polymerization reaction. Some also have an additional chemical-cure reaction and can set in the absence of light.

RMGI restorative materials are indicated for root caries lesions due to their excellent adhesion to tooth structure and fluoride release. They are also recommended for repair of recurrent caries around existing indirect restorations. (Removal of the existing restoration is generally preferred in these situations, but is not always practical.) RMGI restorative materials are the materials of choice for the restoration of NCCLs where esthetics is not of primary importance. RMGI materials have an excellent clinical history of retention with NCCLs, due to excellent adhesion and low elastic modulus, but the long-term esthetic result is compromised. RMGI restorative materials are contraindicated in stress-bearing situations.

When placing RMGI restorations, the following technique is suggested. First, obtain the appropriate shade match. Then obtain the appropriate isolation with a retraction cord/cotton roll or preferably rubber dam. Complete the cavity preparation and then clean the preparation with the dedicated conditioner. Conditioners are formulations of 10%–20% polyacrylic acid, which clean the smeared layer of dentin without opening the dentinal tubules and removing the smear plug. Follow the manufacturer's directions carefully. Some conditioners are designed to be washed off prior to placing the restorative material, while others are not. Mix the material. Some materials are mixed in a triturator (automixed), some are powder/liquid and some are dispensed with a clicker. With Ketac-Nano, do not attempt to place the material until 30–40 s after mixing, as it is initially very sticky and difficult to handle. Place and shape the material and light-cure for the recommended time, typically 20–40 s. Finish and polish. Placement of a surface glaze material is optional.

Cavity Liners

GI liners are recommended when excavation of caries results in a remaining thickness of dentin that is likely less than 2 mm. They are optional any time the depth of a cavity preparation exceeds the depth of a 'minimal' preparation.

Manipulation of GI cavity liners is straightforward. Complete the cavity preparation and clean and dry the cavity. Do *not* desiccate. Mix the lining material and place into the

preparation in a thin layer (no more than 0.5 mm) using a Dycal applicator or similar instrument. Light-cure for 20 s. Now regular bonding procedures are completed when placing composite resin restorations, or Gluma Desensitizer or G5 is applied if placing an amalgam restoration.

There are several GI materials that have been specifically formulated to serve as liners. Among the most popular are Vitrebond and Vitrebond Plus. Both of these materials are light cured, and the difference between them is that Vitrebond is a powder/liquid material and Vitrebond Plus is dispensed with a 'clicker'. Liners are materials that are used at a thickness of no more than 0.5 mm under either amalgam or composite restorative materials. Their function is to seal deep dentin exposed during caries removal with the goal of having about 2 mm of thickness of cavity liner and remaining dentin between the restorative material and the pulp.

A note on fluoride release and secondary caries prevention

Dentists understand that fluoride is an effective caries reducing agent and manufacturers understand that dentists will buy any material that contains fluoride. It is critical to understand that the effectiveness of fluoride as a caries preventive agent is *dose related*. Fluoride releasing amalgam or composite resin has no chance of reducing caries because the dose of released fluoride is so low.

GI–based materials are unique in that the set material consists of an aqueous polygel that permits ion exchange in the oral cavity. This allows GI–based restorative materials to absorb topically applied fluoride from toothpastes, gels, etc., and slowly re-release that fluoride over 24 h.

An almost universal misconception is that all GI materials release fluoride and that this released fluoride is effective in preventing secondary caries. As noted in the introduction, all GI materials do release fluoride. There are two phases to this fluoride release.

- The first occurs primarily in the first 24 h after mixing, and is described as a 'burst' of fluoride release resulting from dissolution of the glass particles in the material by polyacrylic acid.
- After this 24-h period, there is a sustained fluoride release over time, but this level of fluoride release is likely far below the therapeutic level essential to prevent secondary caries.

To reach a potentially therapeutic level, it is essential that GI materials are regularly (3 × /day) exposed to sources of fluoride (e.g. toothpastes, gels, mouth rinses), which will allow the GI material to uptake fluoride and slowly re-release it into the surrounding environment.[16] Research has demonstrated that this does occur on a predictable long-term basis.

Repeated exposure is essential if the GI material is to act as a *reservoir for fluoride*. It is unlikely that a GI-based luting cement, in a 25-µm film thickness under a properly fabricated indirect restoration, would be able to absorb and re-release fluoride at a therapeutic level. A GI base or liner under a sealed restoration would not be exposed to fluoride to be absorbed and re-released. Similarly, under a leaking restoration, these materials would not be exposed to sufficient fluoride to be effective against recurrent caries. The reason these materials are preferred as bases and cavity liners is because of their predictable bond and seal of the underlying dentin, not because of their ability to prevent secondary caries.

Thus, the only GI materials that have legitimate potential for prevention of secondary caries are GI restorative materials such as Fuji II LC, Ketac Nano and Fuji IX. It is important to understand that these materials will only be effective if they receive regular exposures to fluoride (multiple exposures/day).

Clinical Notes

I. GI materials are useful for several clinical applications in restorative dentistry.
II. GI materials provide predictable long-term bonds to enamel and dentin.
III. RMGI cement is the most popular cement in restorative dentistry and is indicated for cementation of metal castings and zirconia and porcelain-fused-to-metal-based restorations.
IV. RMGI liners have an excellent track record as lining materials used in films no thicker than 0.5 mm.
V. RMGI lining cements should be used to provide a liner/dentin thickness of 2 mm from the pulp.
VI. RMGI restorative materials provide satisfactory service for the restoration of root caries and non-carious cervical lesions.
VII. Reinforced GI materials should not be used for core build-ups.
VIII. GI materials do not provide protection from secondary caries with their inherent fluoride release.
IX. GI and RMGI materials may provide protection from secondary caries as a result of their ability to act as fluoride reservoirs. In order to accomplish this, they must be regularly exposed to topical fluoride from toothpastes, gels, mouthwashes, etc.

Adhesive Dentistry

This section is discussed in detail in Chapter 15

Indirect Restorative Materials

Impression Materials

When fabricating indirect tooth-supported or implant-supported restorations, accuracy of the impression is critical. Impression materials have improved steadily during

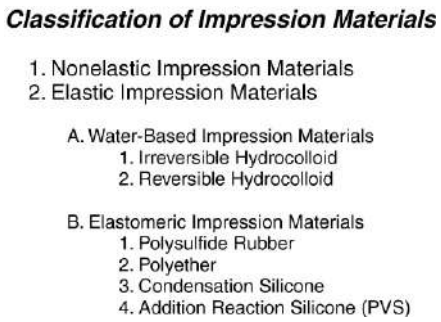


Figure 6.25 Classification of impression materials.

50 years, and the contemporary clinician has a great number of excellent impression materials to choose from.

Classification of Impression Materials

There are two main types of impression materials: nonelastic impression materials, which are used primarily in removable prosthodontics, and elastic impression materials (Fig. 6.25). There are two types of elastic impression materials, water-based and elastomeric (rubberlike) impression materials. The two types of water-based impression materials are reversible and irreversible hydrocolloid (alginate). There are four types of elastomeric impression materials:

I. Addition reaction silicone [also known as polyvinyl siloxane (PVS) impression material]
II. Condensation silicone
III. Polysulphide rubber (also known as rubber base, thiokol rubber and mercaptan rubber)
IV. Polyether

Ideal Properties of Impression Materials

Ideal properties to be discussed include accuracy, elastic recovery, dimensional stability, flow, flexibility, workability, hydrophilicity, shelf life, patient comfort and economics.

Accuracy

There are three main factors related to accuracy of impression materials:

i. Fine detail reproduction

- According to ADA #19, elastomeric impression materials used to fabricate precision castings must be able to reproduce *fine detail 25 μ or less*. All currently available impression materials meet this specification.
- Polyvinyl siloxane is the best in this regard and reversible hydrocolloid is the worst. It is not likely that improved ability to record fine detail results in improved accuracy of the final restoration, because the limiting factor in this area is the ability of the gypsum die materials to record fine detail. The corresponding specification for gypsum die materials related to fine detail is 50 μ. Some die materials can provide better fine detail reproduction than the 50-μ specification but fall well short of the impression materials in this regard.
- There are differences in the ability of different viscosities of elastomeric impression materials to record fine detail. Typically, the *lower the viscosity, the better the ability to record fine detail*. Putty viscosities in general are very poor in this regard, and are only required to record fine detail of 75 μ.[17]

ii. Dimensional accuracy

- A second aspect is dimensional accuracy, which is evaluated by measuring tooth-to-tooth distances in the same quadrant and also cross-arch distances.
- All of the available contemporary impression materials provide superb accuracy if they are manipulated correctly. Although PVS materials are likely to be slightly more accurate than other impression materials, differences in accuracy are not likely to be clinically significant.

iii. Elastic recovery

- Elastic recovery is the ability to flow into an undercut, set in that position, and then spring around the undercut on removal of the impression and rebound to its original shape.
- Polyvinyl siloxane impression materials have the best elastic recovery at over 99% with a specific test undercut. This elastic recovery coupled with excellent dimensional stability make it the material of choice for clinicians who prefer to double pour impressions to obtain a backup die.
- No impression material has 100% elastic recovery; and for all impression materials, the greater the depth of the undercut, the greater the permanent distortion of the impression material is.[18,19]
- At one time, dentists were cautioned to wait 30 min prior to pouring impressions to allow maximum elastic recovery to occur. With contemporary impression materials, *this delay is no longer necessary* as elastic recovery occurs almost instantly.
- Keeping in mind the greater the depth of the undercut, the greater the distortion, an important procedure is to always block out undercuts in tooth preparations prior to impression making. This often occurs when preparing teeth with existing restorations. Many dentists fail to do this, assuming the laboratory can block out the undercuts on the die. While this is possible, blocking out the undercuts with RMGI materials will reduce the overall distortion due to improved elastic recovery.

Dimensional Stability

One of the most important properties when selecting an impression material is dimensional stability. An ideal impression material would be dimensionally stable over time, allowing the impression to be poured at the convenience of the operator. This is extremely important in contemporary dentistry, as most dentists have opted to not pour up their own impressions and instead send them to

dental laboratories to be poured. The time between removal of an impression from the mouth and the time when it is poured by the laboratory varies considerably and if the impressions are mailed to the lab, it will be well over 24 h.

Dimensional stability of an impression material is determined by:

- Composition of the material
- Its susceptibility to imbibition and syneresis
- Volatility of the by-product of the setting reaction.[20]
  - i. ***Water-based impression materials:*** Impressions made from water-based impression materials such as reversible and irreversible hydrocolloid (alginate), are approximately 80% water, which evaporates at room temperature. This evaporation can result in clinically significant distortion of the impression. Because of this evaporation (syneresis), *water-based impression materials should be poured within 10 min* after removal from the mouth. Water-based impression materials are also subject to imbibition or the absorption of water. The common practice of wrapping alginate impressions with wet paper towels should be avoided, as the impression material will absorb water from the towel (imbibition), which results in distortion of the impression. These impressions are properly stored in sealed plastic bags prior to pouring.
  - ii. ***Condensation silicone impression material:*** The by-product of the setting reaction of condensation silicone impression materials is *ethyl alcohol*, which evaporates rapidly at room temperature. Impressions made with *condensation silicone should be poured within 30 min* of removal from the mouth.
  - iii. ***Polysulphide impression material:*** The by-product of the setting reaction of polysulphide rubber (rubber base) impression materials is *water* (Fig. 6.26). This evaporates more slowly than it does from water-based impression materials, but is still significant. Thus, *impressions made with polysulphide rubber should be poured within 30 min*.
  - iv. ***Polyether impression material:*** Polyether impression materials have no volatile by-product of the setting reaction, but are sufficiently hydrophilic to absorb water from the atmosphere. This absorption is dependent on relative humidity (RH); and if the RH is over 50%, absorption of water is sufficient to cause clinically significant distortion.[21] *Polyether impressions should be poured within 1 h* and should never be mailed to a dental laboratory.
  - v. ***PVS impression material:*** There is no by-product of the setting reaction for PVS impression materials, so *PVS impressions possess almost unlimited dimensional stability*. If impressions are to be mailed to the dental laboratory, the only reasonable material to use is PVS. Indeed, the case can be made that if the impression is going to be poured by a local laboratory, it is safer to use PVS because there is no way to be certain it will be poured at the appropriate time. *This exceptional dimensional stability, coupled with the excellent elastic recovery of PVS impression materials, make it the best choice for clinicians who prefer to double pour impressions to obtain a backup die*.

Table 6.10 displays discussed impression materials and recommended pouring times.



Figure 6.26 Rubber base impression material.

Flow and flexibility

Impression materials need to flow readily into the minute details of cavity preparations and accurately capture grooves, pinholes and cervical margin detail. Most brands of impression material provide *light body* or syringe materials with low viscosity to accomplish this. These are used with *heavy body* materials in a tray to provide more rigidity to the impression and assist in forcing the low viscosity material into the gingival sulcus and accurately and completely capture the prepared cervical margin.

Set impression materials vary considerably from one another with regard to flexibility.

- Polyether impression materials tend to be more rigid than other materials. This can present a problem with long, thin preparations of mandibular incisors or

Table 6.10 Multiple impression materials and recommended pouring time

Impression material	Pour time
Water based	10 min
Polysulfide rubber	30 min
Condensation silicone	30 min
Polyether rubber	60 min
Polyvinyl siloxane	At operator convenience

periodontally involved teeth. Fracture of the delicate gypsum dies is a common occurrence due to the rigidity of polyether materials. Another problem related to this rigidity is tearing of the impression material in the gingival sulcus. While the tear strength of polyether impression materials is excellent, the amount of force required to remove the rigid impression may exceed the tear strength of the material.

- Although rarely used today, reversible hydrocolloid impression material is the least rigid of all impression materials and may be the material of choice when making impressions of multiple prepared periodontally involved teeth.

With dual-arch impressions, it is advantageous to use a relatively rigid material. Many of the commonly used trays for dual-arch impressions are somewhat flexible, and a rigid impression material can compensate for this flexibility. Many PVS products have heavy body versions that provide the essential rigidity.

Workability

Impression materials were originally supplied as two tubes for each viscosity. One tube was labelled as the *base*, the other the *catalyst* (Fig. 6.27). The base and the catalyst were different in colours and equal lengths of each were dispensed on a mixing pad. The materials were mixed until the material became a uniform colour with no streaks.

When making a clinical impression, one pad with heavy body material was used and one with light body material. The materials were mixed simultaneously, the light body material loaded into an impression syringe and the heavy body material loaded into a tray. This process required excellent coordination between the dentist and assistant(s); thus impression making was a stressful procedure.

Contemporary materials are supplied with 'automix' handheld devices. This is a substantial improvement and results in a standardized mix with fewer inherent porosities, increased working time and economic savings due to less waste of material. Even better are sophisticated electronic mixing devices such as the Pentamix (3M Oral Care, St. Paul, MN) (Fig. 6.28).



Figure 6.27 Two-tube impression material: base and catalyst.



Figure 6.28 Electronic impression mixing device. (Courtesy 3M Oral Care, St. Paul, MN)

Working times of impression materials vary by the manufacturer, and most automix devices provide materials with setting capabilities of either:

I. Standard-set: When making a single crown impression with a quadrant or dual-arch tray, the operator may opt for a quick-set material with a shorter working time.
II. Quick-set: When making a full-arch impression with multiple prepared teeth, it might be prudent to choose a material with a longer working time. When making an impression with several prepared teeth, another option to extend working time is to refrigerate the low viscosity material, which increases working time without sacrificing accuracy.[22]

Hydrophilicity

Reversible hydrocolloid impression materials are truly hydrophilic and can effectively make accurate impressions in the presence of moisture.

- ***All elastomeric impression materials require a completely dry field to make a successful impression.***
- Polyether impression materials are hydrophilic as witnessed by their tendency to absorb water from the atmosphere. However, polyether impression materials absolutely require a dry preparation surface to make an accurate impression.
- The manufacturers of many contemporary PVS impression materials claim their products are 'hydrophilic', and incorrectly imply that their materials can effectively make impression in the presence of blood and/or

saliva. The hydrophilicity of a material is measured by the contact angle a sessile drop makes with the material. If the contact angle is below a specific value, the material can be called hydrophilic. Many contemporary PVS impression materials meet that criterion, and therefore the manufacturers are technically correct. However, the implication that these materials can make impressions in a wet environment is totally incorrect.[23,24] *PVS impression materials require a dry environment to make successful impression.*

The advantage these 'hydrophilic' PVS materials have is they are significantly easier to pour. Early versions of PVS were hydrophobic and very difficult to pour. They also released hydrogen for approximately 3 h after setting, so delaying pouring for that period was recommended. Contemporary PVS materials have non-ionic surfactants grafted onto the polymer chains, which makes them hydrophilic and thus easier to pour. They also contain small quantities of spongy palladium, which absorbs hydrogen and allows pouring of the impression immediately upon removal from the mouth.

Shelf life

The exact shelf life of impression materials is not known, but it is inadvisable to use impression materials that have passed the expiration date established by the manufacturer. Dentists must become familiar with manufacturers' codes for the materials they are using and ensure that current materials are consistently utilized. It is arbitrarily suggested that no more than a 6-month supply of impression material should be kept on hand at any time.

Patient comfort

Contemporary impression materials are far more patient friendly than the old polysulphide materials, which were messy, had a bad taste and odour and permanently stained clothing. Reversible hydrocolloid materials were uncomfortable for patients because they required the use of bulky, water-cooled trays. Contemporary impression materials are essentially odourless and tasteless.

The rigidity of polyether impression materials can be a disadvantage, especially if the patient has existing fixed prostheses or multiple open gingival embrasures due to periodontal disease. In these situations, it is advisable to use a more flexible material, and to block out the undercuts with utility wax prior to impression making.

The ultimate in patient comfort with impression making is the use of optical impressions. Digital impressions are becoming more and more popular and are very patient friendly. Imaging systems are becoming more economical and it is anticipated that the majority of impressions made in the next decade will be made optically. It is important to point out that while optical impressions for single tooth preparation are as accurate as conventional impressions, the same factors that result in failed contemporary impressions will result in failed optical impressions.

Economic factors

There can be significant differences between the costs of different impression materials. Reversible hydrocolloid impression material is considerably less expensive than elastomeric impression materials. However, there is a significant cost involved with using reversible hydrocolloid, considering conditioning and tempering baths need to be purchased along with an appropriate selection of water-cooled stock trays. Polyether and PVS materials are comparable in costs.

Clinical Notes

I. Many excellent impression materials are available to the contemporary clinician.
II. The quality of impressions sent to commercial dental laboratories needs improvement.
III. All contemporary impression materials are sufficiently accurate if manipulated properly.
IV. The only impression material that should be mailed to commercial laboratories is PVS.
V. Polyvinyl siloxane materials have the best fine detail reproduction, elastic recovery and dimensional stability of all impression materials.

Principles of Impression Material Manipulation

Proper manipulation of impression materials is probably more important in determining the accuracy of an impression than which type of material is selected. This section will discuss the important manipulative variables that must be considered when making impressions for indirect restorations.

Control of Bulk

Impression materials shrink when setting. Irreversible hydrocolloid and elastomers shrink due to polymerization reactions, and reversible hydrocolloid undergoes thermoplastic shrinkage as it cools from the warm to cool state. The effect of this shrinkage can be controlled by paying close attention to the bulk of impression material used. *Water-based impression materials are most accurate when used with a uniform cross-sectional thickness of 4–6 mm.* This means that successful impressions can be made using stock trays that will provide approximately 4- to 6-mm thickness of material.

Elastomeric impression materials are most accurate when used with a cross-sectional thickness of 1–2 mm.[15] Use of a stock tray with elastomers results in excess thickness of impression material, which possibly could result in an inaccurate impression. Elastomeric full-arch impressions are best made using a custom tray that will provide the desired cross-sectional thickness of impression material (Fig. 6.29). Numerous studies have evaluated the accuracy

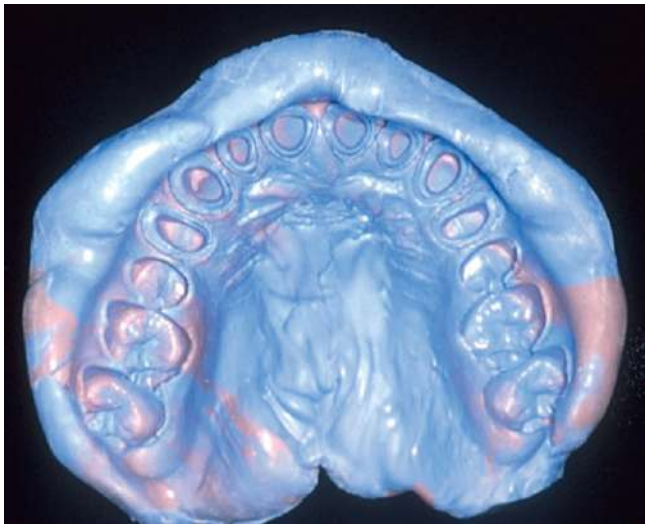


Figure 6.29 Full-arch impression taken with a custom tray.

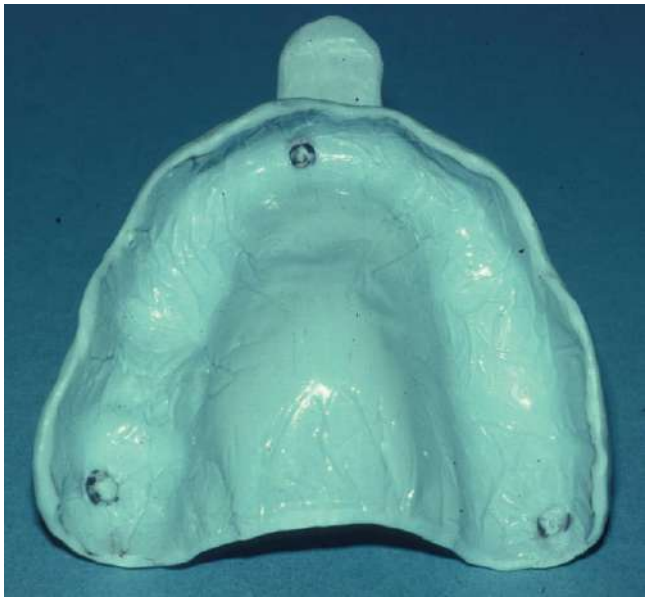


Figure 6.30 Custom tray showing three occlusal stops.

of elastomeric impressions made in stock trays versus impressions made in custom trays. Almost all conclude that impressions made in custom trays are more accurate, yet very few dentists routinely use custom trays.[25]

The difference in cross-sectional thickness of impression material in a stock tray versus a custom tray is only about 2 mm, but that difference does reduce the accuracy of the impression.[26] This mandates that custom trays be fabricated with precision.

Fabrication of Custom Tray:

- Custom trays should be fabricated on the diagnostic cast using *one layer of base plate wax as a spacer*.
- Trays can be fabricated with PMMA, photo-cure bisacryl (Triad, DENTSPLY Sirona, York, PA), or PVS. PMMA trays should be fabricated at least 24 h in advance to ensure stability.
- Occlusal stops are critical for proper orientation of the tray in the patient's mouth. Three occlusal stops are ideal, with at least one stop posterior to the prepared teeth. Stops should be placed on nonfunctioning cusp tips to minimize distortion in the area of the stops. The stops are prepared by removing the base plate wax from the nonfunctioning cusp tips with a hand instruments (Fig. 6.30).
- The wax spacer should be covered with tin foil before making the tray to facilitate removal of the wax from the tray and to prevent incorporation of a wax residue on the internal surface of the tray. This can happen due to the inherent exothermic reaction that occurs during the setting of the PMMA tray material. This residue can interfere with the proper functioning of tray adhesives.

Adhesion of the impression material to the tray

It is imperative that the impression material adheres to the tray. With proper adherence, the impression material shrinks ***toward*** the tray as it polymerizes, creating a slightly larger die. Adhesion is achieved using specific chemical tray adhesives. Adhesives must be specifically matched with the impression material. They should be painted in a thin layer on the internal portion of the tray and the tray borders (Fig. 6.31). The tray should be painted 7–15 min prior to impression making to permit formation of an adequate bond strength of the impression material to the tray.[27]

Viscosity control

Impression materials are supplied in a number of viscosities ranging from very low viscosity to very high viscosity putty materials. The main difference between these materials with different viscosities is in the amount of *inert filler material* in the impression material. With low viscosity materials, there is relatively little inert filler and thus more chemical reagent. Low viscosity materials are excellent at recording fine detail, but also undergo more polymerization shrinkage during the setting reaction.

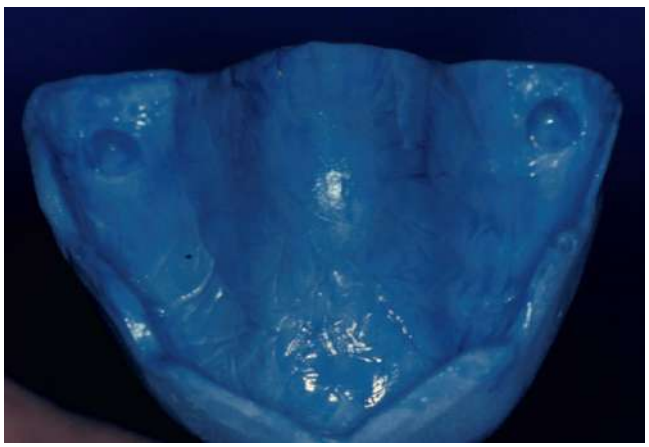


Figure 6.31 Custom tray coated with tray adhesive.

Optimum method of impression making:

- Use only as much low viscosity material as is necessary to capture the fine detail of the prepared margin, grooves and box forms of the preparation; and the bulk of the impression should be made with high viscosity material.
- This high viscosity material has correspondingly more inert filler and less reagent and displays less polymerization shrinkage on setting.
- The heavy body material also assists in pushing the low viscosity, light body material into the gingival sulcus.

Monophase PVS and polyether materials are supplied by many manufacturers. In theory, such materials do not provide the same level of accuracy as provided by the proper use of a combination of low viscosity/high viscosity materials, but the actual differences in accuracy are likely not clinically significant. The convenience of having to use only one viscosity of material makes monophasic materials very practical.

Adequate mixing

As discussed in the previous section, almost all impression materials today come with an *automix system*. This provides optimum mixing with fewer inherent voids, extends the essential working time of the materials and reduces waste because the materials are loaded from the dispenser directly into the syringe or tray.

Disinfection

Microorganisms in the oral cavity can be transmitted from impressions to the dental laboratory. Dentists must disinfect impressions prior to pouring the cast or sending the impression to the laboratory.

I. The first step of any disinfection technique is to rinse the impression thoroughly in cold tap water. This step removes a significant portion of the microorganisms from the impression.
II. Disinfection techniques involve spraying the impression with disinfection agents or immersion of the impression materials in chemical agents such as sodium hypochlorite.
III. Polyvinyl siloxane materials are stable in this regard, but special care must be taken with water-based materials and polyethers to ensure that adequate immersion times are used to eliminate microorganisms but that extended immersion times are avoided to prevent excess imbibition of the disinfecting solution and distortion of the impression.

Pouring of impression materials

One of the most important manipulative variables with impression materials is the time limit after removal from the mouth to when the impression is poured with the appropriate gypsum product.[28] The factors that determine when an impression should be poured are discussed in depth in the previous section (Table 6.10).

Polyvinyl siloxane impression materials and latex

Polyvinyl siloxane impression materials are easily the best selling impression materials in the market, and they have the best fine detail reproduction and the best elastic recovery of all available impression materials. Polyvinyl siloxane materials have remarkable dimensional stability and are odourless and tasteless and pleasant for patients. They are provided in a wide variety of viscosities, rigidities and working and setting times so they can be used in a variety of clinical situations.

Polyvinyl siloxane impression materials have one major disadvantage: they have a significant interaction with latex (rubber dam and latex gloves). Any contact of unpolymerized PVS material with latex results in inhibition of polymerization of the impression material. This can occur if the dentist or assistant mixes putty materials while wearing latex gloves or even if latex gloves were worn prior to mixing.

- *Direct inhibition* of polymerization also can occur if the impression material is in contact with a rubber dam.
- *Indirect inhibition* of polymerization also can occur intraorally when latex gloves contact tooth preparations and the surrounding periodontal tissues during tooth preparation or gingival displacement procedures.[29–32] Such inhibition of polymerization is often subtle and limited to small isolated areas of the surface of the impression. It is often not detected with the initial inspection of the impression and may be noticed only after pouring and separation of the gypsum cast.

The presenting signs of inhibited polymerization are a film of unset material in isolated areas or the presence of a sticky, slippery substance on the surface of the impression (Fig. 6.32). It is similar to the feel and appearance of the oxygen-inhibited layer that is seen with photo-cure composite resin restorative materials.

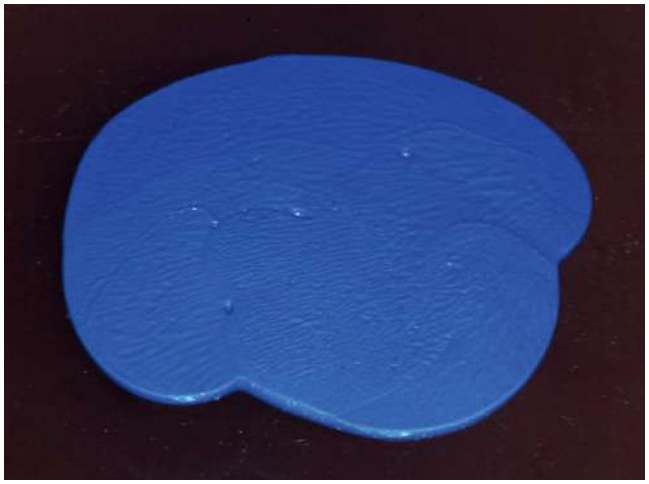


Figure 6.32 Unpolymerized polyvinyl siloxcane material due to handling of the material with latex gloves.

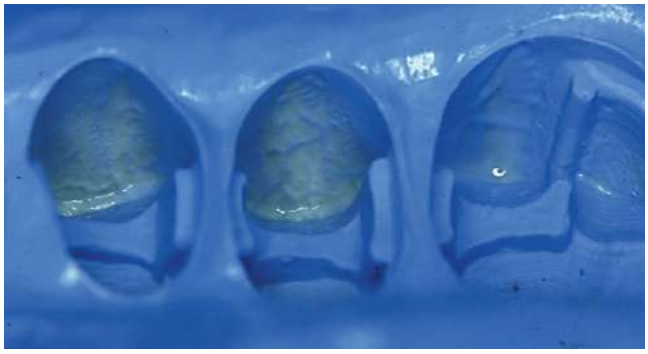


Figure 6.33 Impression displaying areas of inhibited polymerization of the prepared teeth.

Although these isolated areas of inhibited polymerization are subtle and not easy to detect (Fig. 6.33), depending upon their location they can render an impression unusable. Clinicians must inspect impressions and recovered casts carefully to ensure contamination of critical areas has not occurred.

The mechanism of inhibition of polymerization is thought to be a result of contamination of the chloroplatinic acid catalyst of the PVS material with unreacted sulphur, which is present in natural latex gloves. Synthetic latex gloves, vinyl gloves and the powder commonly found on gloves do not cause this inhibition of polymerization. Sulphur-containing gingival displacement chemicals do ***not*** cause inhibition of polymerization.[33]

Clinicians should avoid touching tooth preparations and adjacent gingival areas with latex gloves. When this is not avoidable, wearing vinyl gloves over the latex gloves is recommended. Once contamination of the preparations has occurred, cleaning with water and other agents is not effective in removing the contamination. Routine cleansing of tooth preparations with flour of pumice may be indicated prior to impression making.

Contemporary PVS impression materials may not be as sensitive to latex as the products that were used in the 1990s. It does appear that direct contact of contemporary PVS with latex or nitrile gloves will result in inhibited polymerization; however, these materials are not susceptible to indirect contamination.[34] It appears this reduced sensitivity is a result of changes to the impression materials rather than changes in gloves and rubber dam materials. Contemporary PVS materials have increased concentrations of platinum in the chloroplatinic acid catalyst, which makes it less sensitive to sulphur. Also non-ionic surfactants have been grafted onto the polymer chains and these surfactants may act as a separating medium and prevent the sulphur from contacting the catalyst.

Putty-wash impression techniques

While the recommended technique for full-arch impressions is use of heavy body/light body viscosities of impression material in a custom tray, many dentists prefer to make 'putty-wash' impressions.[20] There are three approaches to putty-wash impressions. One approach is appropriate and acceptable, another can provide acceptable impressions but has some potential drawbacks and a third approach is unacceptable.

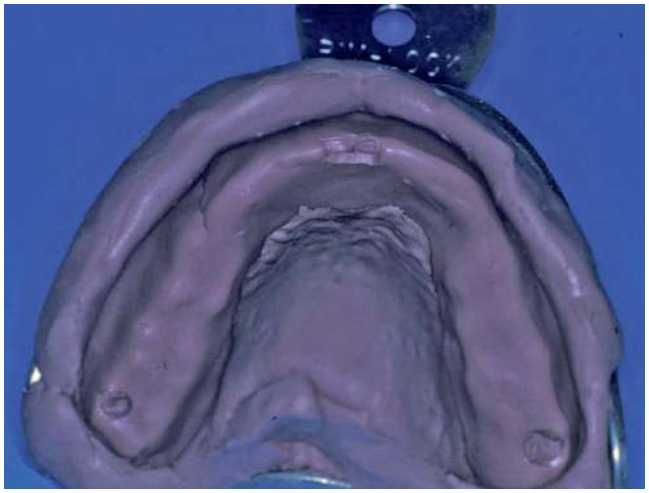


Figure 6.34 Polyvinyl siloxane putty custom tray.

I. *Custom tray putty-wash impression technique*: An excellent technique for putty-wash impressions is to use the putty material to fabricate a custom tray. It is fabricated in the same manner as with PMMA materials or light cure materials. One layer of base plate wax is placed over the diagnostic cast as a spacer, and wax is removed from nonfunctioning cusp tips to provide occlusal stops. A putty impression is made in a stock tray, producing a PVS putty custom tray (Fig. 6.34). Now the light body viscosity impression material is syringed on to the PVS putty custom tray and the impression is made.

II. *Relieved putty wash impression technique*: In this technique, a preoperative putty impression is made intraorally prior to tooth preparation. Plastic sheets may be placed over the teeth to prevent putty material from entering gingival embrasures. In the area where the teeth are to be prepared, putty impression material is removed with a bur or scalpel to provide relief, and the impression is 'washed' or relined with low viscosity PVS material.
This approach can be successful, but there are two potential problems.
   i. It is difficult to confine the wash materials to the area of the relieved impression, and some wash material enters the unrelieved impression. This results in an inaccurate occlusal pattern for the resultant cast.
   ii. Thus, the entire impression rather than just the relieved area should be 'washed'. This creates the problem of hydraulic distortion of the putty material as the impression is seated in the mouth. This is impossible to detect on a clinical level but may have a deleterious effect on the accuracy of the impression and resulting restoration.

III. *Simultaneous* putty-wash impressions *technique*: With this technique a stock tray is loaded with putty material,

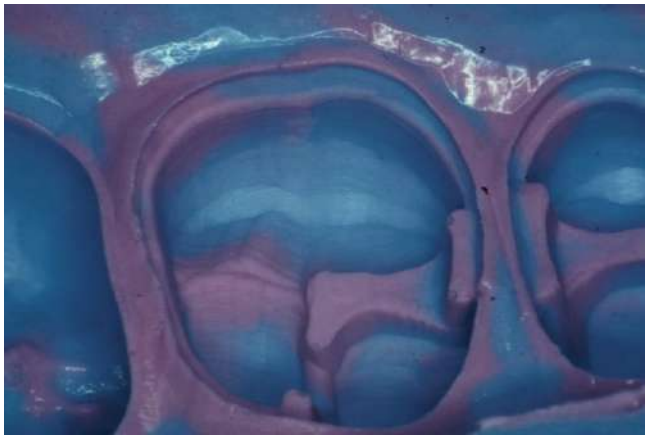


Figure 6.35 Impression taken utilizing the 'simultaneous technique'. Notice portions of the margin captured with putty material, which may result in lack of details.

and the syringe material is injected around the prepared tooth or teeth. The tray containing the putty material is squashed over the syringe material, and the impression is made with the putty and syringe material setting simultaneously. This approach is unacceptable because:

i. It is impossible to control the thickness of the impression material, and excess bulk is used.
ii. It is impossible to control what material records the margin detail of the preparation(s). Usually, portions of the prepared margin are captured in the putty, and putty materials are essentially deficient in their ability to record marginal detail (Fig. 6.35).

Clinical Notes

I. Water-based impression materials are most accurate with a uniform cross-sectional thickness of 4–6 mm. This allows impressions made with these materials to be made in a stock tray.
II. Elastomeric impression materials are most accurate when made with a uniform thickness of 1–2 mm. This implies the routine use of custom trays.
III. The rational for custom trays is to make impressions with a uniform cross-sectional thickness of 1–2 mm.
IV. Custom trays can be made with PMMA, putty and photo-cure bisacryl.
V. Occlusal stops should be placed on nonfunctioning cusp tips.
VI. Low viscosity materials should be used only around the prepared teeth.
VII. Specific impression adhesives should be used and applied to custom trays 7–15 min prior to making the impression.
VIII. Impressions should be poured at appropriate times for each type of material.
IX. Impressions should be disinfected prior to pouring or sending to the laboratory.
X. Some putty-wash techniques are superior to others.

Casting Investments

The lost wax process was introduced to the profession in 1907 by Taggart.[35] In order to fabricate a precise gold casting, it is necessary to balance a number of expansions and contractions related to dental materials.

One of the major shrinkage factors in the process is the casting shrinkage of the metal alloys as they cool from the liquidus temperature to bench temperature. This shrinkage is roughly proportional to melting temperature: the higher the melting temperature the greater the shrinkage of the alloy.

- Type III gold alloys shrink approximately 1.2%
- Base metal alloys have double the casting shrinkage at 2.4%.

This shrinkage must be compensated in the entire process. The major mechanism for accomplishing this is the use of casting investments.

There are three major types of casting investments:

I. **Gypsum-bonded investments** are primarily used with gold casting alloys.

With gypsum-bonded investments there are three major components.

i. The **refractory** is composed of materials such as cristobalite, trydimite and quartz. The refractory supplies most of the thermal expansion of the investment through the process of **inversion.** As the investment is heated in a burnout oven, the investment expands in proportion to the increase in temperature. Then a significant amount of expansion suddenly occurs when a specific temperature, called the inversion temperature, is reached. This expansion occurs as a result of a change in crystalline form of the refractory; it changes from the alpha crystalline form to the beta crystalline form. This form change is associated with specific expansion called inversion.
ii. The second component of gypsum-bonded investments is the **binder.** The binder is basically gypsum, starting out as calcium sulphate hemihydrate powder converting to calcium–sulphate dihydrate after being mixed with water. The setting reaction of gypsum is associated with expansion, which contributes along with the thermal expansion to compensate the casting shrinkage of the alloy.
iii. The third component of gypsum-bonded investments is the **modifier**, which may take the form of oxidizing agent or reducing agent, depending on the commercial product.

There are three types of expansion associated with gypsum-bonded investments.

– The first and most important is **thermal expansion** as a result of refractory expansion, as described previously.
– The second is **setting expansion** provided by the setting reaction of the gypsum binder.
– The third type is **hygroscopic expansion** provided by soaking the ceramic liner of the casting ring. At

one time, asbestos was used for this function. The liner allows the investment and casting to be easily removed from the casting ring after casting is completed. By soaking it in water before placing it into the ring, the liner supplies a small amount of gagging water to the setting gypsum, which in turn allows a small amount of extra expansion.

Note: There is a separate type of investment called a hygroscopic investment. These are very accurate investments that rely primarily on hygroscopic expansion and will not be described here because they are rarely used at this time.

II. **Phosphate-bonded investments are** also known as high-heat investments. They are primarily used with porcelain bonding alloys because they can withstand the higher temperatures required to melt porcelain-bonding alloys. Phosphate bonded investments were developed to provide the extra expansion required with metal alloys that melt at high temperatures and also had to be formulated to withstand those higher melting temperatures. Phosphate-bonded investments come with a 'special liquid', which is basically liquid silica. The liquid replaces some of the water used to mix the investment, and the basic rule is the more liquid silica, the greater the expansion of the investment.

III. **Ethyl-silicate investments** are primarily used with removable partial denture castings. Manipulation of casting investments is relatively simple. First the appropriate investment materials must be selected. Then the most important variable is use of precise water to powder ratios. In the case of phosphate-bonded investments, liquid to powder ratios are used, with the liquid being part water and part special liquid, depending on the amount of expansion required.

Investments are vacuum mixed and poured into different sized moulds, depending on the size of the casting. After waiting a minimum of 1 h, the casting ring is placed into a cold oven, subjected to a controlled burnout and the restoration is cast.

Clinical Notes

I. Casting investments are an integral part of the lost wax process.
II. The three types of casting investments are gypsum bonded, phosphate bonded, and ethyl silicate investments.
III. Gypsum-bonded investments are used with cast gold; phosphate-bonded investments are used with porcelain bonding alloys, and ethyl-silicate investments with removable partial denture frameworks.
IV. Casting investments compensate for casting shrinkage of the alloy.
V. Three types of expansion occur with casting investments: thermal, setting and hygroscopic. The most important expansion is thermal.
VI. Special liquid (liquid silica) is used with phosphate-bonded investments to allow for the precise amount of expansion depending on the alloy used.

Gold Alloys

Cast gold restorations have been used in dentistry ever since Taggart introduced the lost wax process in 1907.[35] Properly done cast gold restorations provide the longest term survival of all types of restorations. A recent study of 1314 cast gold restorations in place from 1 to 55 years demonstrated that at 30 years, the failure rate was 4.5% (Fig. 6.36).[36]

Gold is used because it is a **noble** metal and does not tarnish in the oral cavity. All noble metals share that property. The major noble metals used in dentistry are *gold*, *platinum* and *palladium*. Nobility is determined by an element's position in the periodic table. Silver is a precious metal, but is not noble as it easily tarnishes in the oral cavity.

Pure (100%) gold is too soft for use in the oral cavity, so it must be alloyed with other metals to produce the essential physical properties. Metals that are alloyed with gold in dental casting alloys include:

I. platinum and/or palladium for both strength and passivity
II. silver and copper for strength
III. zinc and indium as deoxidizers

There are four types of gold alloy, Types I, II, III and IV (Table 6.11).



Figure 6.36 Cast gold restorations after 37 years.

Table 6.11 Types of gold alloys used in dentistry

Type	Au	Hardness	Indication
I	98%	Soft	Class V
II	77%	↑	Inlays
III	72%		Crowns
IV	69%	Hard	RPDs

I. *Type I gold alloy* has the highest concentration of gold at 98%; it is considered a 'soft' gold and is indicated in non-stress-bearing situations such as Class V inlays.
II. *Type II gold alloy* contains approximately 77% gold and is typically used with gold inlays and restorations with finishable margins, as these alloys can be burnished to attain maximum marginal integrity.
III. *Type III gold alloy*, with a concentration of 72% gold, is the most commonly used gold alloy in dentistry; it is used to fabricate crowns and fixed partial dentures.
IV. *Type IV gold alloy* has the lowest concentration of gold (69%), and was used for the fabrication of removable partial denture frameworks at one time but rarely for that purpose now. It is still used for the fabrication of custom cast dowel and cores, as it is the strongest of the gold alloys and is less expensive due to the lower concentration of gold.

Type I gold is the softest, with the highest concentration of gold; Type IV is the hardest, with the lowest concentration of gold. Types I and IV gold are rarely used at this time, thus the most used alloys in contemporary dentistry are Types II and III.

One technique that can be used to improve marginal integrity of cast gold restorations is *heat treatment*. To heat treat a gold alloy, the alloy must have a copper content of 12%. Metal alloys can exist in one of three states. With a *eutectic mixture*, distinct phases can be identified in the microstructure. A good example of a eutectic would be amalgam where gamma, gamma-1 and gamma-2 phases can easily be identified in a micrograph. Eutectic mixtures are hard and brittle.

The second state seen with metal alloys is as an *intermetallic compound*. With intermetallic compounds there are no distinct phases, but there is molecular ordering, and these alloys are hard and strong. After *heat hardening* treatment, gold alloys exist as intermetallic compounds. When gold alloys are *heat softened*, they exist as a *solid solution*. With solid solutions, there is a total lack of molecular order with randomization, and solid solutions are soft and burnishable.

The heat treatment procedure begins by *heat softening*. The alloy is melted and cast and immediately quenched in cold water. This transforms the alloy into a solid solution, which again is soft and burnishable. The restoration is then taken to the mouth and the finishable margins are burnished to achieve maximum marginal integrity. Then the alloy is *heat hardened* to render it hard and strong. To accomplish this, the restoration is heated in an oven and then slowly cooled to bench temperature. This transforms the alloy into an intermetallic compound, which will be hard and strong and provide optimum clinical service.

The use of cast gold has declined significantly in recent years primarily due to the profession's obsession with esthetics and also due to the cost of gold.

Clinical Notes

Cast gold restorations unquestionably provide the longest service of any restoration available in dentistry

Porcelain Bonding Alloys

In late 1962, metal-ceramic restorations were patented and introduced to the profession by Weinstein, Katz and Weinstein.[37] Original research in this area began with typical gold casting alloys, simply because they had been used in dentistry ever since Taggert had introduced the lost wax process to the profession in 1907. Four major alterations needed to be made to these alloys before clinical success could be expected.

I. The melting temperature needed to be raised because the gold alloys melted at temperatures well below the fusing temperatures of porcelain. This was achieved by the addition of an appropriate amount of platinum to the gold alloy.
II. Strength of the alloy needed to be increased. Any flexure in the metal substructure would result in catastrophic fracture of the porcelain veneer. Conventional gold alloys are far too flexible. Again, the addition of platinum was a major factor in improving the strength of the alloy.
III. The thermal coefficients of expansion and contraction of the metal alloy and the veneering porcelain needed to be more closely matched. The COE of the alloy was reduced to a great extent again by the inclusion of platinum, and the COE of the veneering ceramic was raised with the addition of sodium.
IV. Predictable bonding of the porcelain to the metal needed to be achieved. Mechanisms of porcelain to metal bonding will be dealt with in more depth later on in this chapter. With these early alloys, chemical bonding was achieved with the inclusion of trace elements such as indium, zinc and tin. These elements produced a consistent oxide on the surface of the alloy that would provide a chemical bond with the tetravalent oxides in the opaque porcelains.

These early porcelain-bonding alloys contained a high percentage of gold (85%), and were used extensively through the 1960s and early 1970s. Although the physical properties of these high gold alloys were less than ideal, the main impetus for research into alternative alloys was the sudden increase in the price of gold in the late 1970s. Since that time, a number of alternative alloy systems have been introduced and have been extensively researched. The following analysis is given to assist the clinician in alloy selection for this important clinical modality. It is important to understand that the dentist should determine which alloy should be used for each restoration. This important choice should not be delegated to the laboratory technician.

Classification

Because the physical properties of an alloy can be predicted from their component elements, it is useful to classify porcelain-bonding alloys according to their composition. Often alloys are casually referred to as 'precious', 'semiprecious' or 'nonprecious' metals, usually based on the approximate gold content of the alloy. These terms lack precision, and several alloy groups with significantly different physical properties could be legitimately termed 'semiprecious'.

A more meaningful classification is to use the ***nobility*** of an alloy to designate the appropriate group to which a metal alloy should be assigned. The nobility of an element is determined by its position in the periodic table, and is a measure of its passivity in the oral environment. The noble metals most commonly used in restorative dentistry are *gold*, *platinum*, and *palladium*.

Iridium, rhodium, ruthenium and osmium are also designated as noble metals. Silver, which has a tendency to tarnish in oral fluids, is generally not considered as a noble metal. Silver is a precious metal, which signifies only that the metal has some intrinsic value. *The word precious has nothing to do with tarnish potential.*

Thus, porcelain-bonding alloys can be classified into two major groups, namely noble alloys and base metal alloys.

There are three subgroups of noble alloys and three subgroups of base metal alloys (Table 6.12). They can be differentiated from one another in terms of composition; certain attributes and physical properties such as strength, fit of castings and bond potential with porcelain, biocompatibility and economic considerations. Table 6.13 compares the alloy systems in terms of these variables.

Table 6.12 Classification of metal-porcelain alloys

Noble metal alloys	Base metal alloys
High gold	Nickel-chrome-beryllium
Low gold	Chrome-cobalt
Palladium-silver	Titanium

Composition

I. **Noble alloys**
   i. ***High gold alloys*** typically contain a high percentage of gold, usually in the range of 80%–85%. The alloy is strengthened with the addition of 7%–10% platinum, and trace elements such as indium, zinc and tin are added to provide an oxide layer for predictable porcelain bonding. These alloys predominated until the price of gold skyrocketed in the mid-1970s. Examples of commercial alloys in this category are Jelenko-O (J.F. Jelenko Co., Armonk, NY), Silhouette 850SL (Leach and Dillon, N. Attleboro, MA) and SMG-3 (J.M. Ney Co., Bloomfield, CT).
   ii. ***High noble alloys*** have a gold content typically in the range of 50%–55% gold. Palladium is present in the range of 35%–40%, along with the trace elements essential for bonding. These alloys may or may not contain silver. If more than 12% silver is present, it may cause a tendency for the porcelain to take on a greenish hue. Small amounts of silver may improve the wettability of the metal coping with the opaque porcelain, although this has not been scientifically established. Examples of this type of alloy include Olympia (J.F. Jelenko, Armonk, NY), USC Ceramic Alloy (Leach and Dillon, N. Attleboro, MA) and W2 and W3 (Williams Gold Co., Buffalo, NY).
   iii. ***Palladium–silver alloys*** are composed primarily of high concentrations of palladium (60%) and silver (30%). With such high concentrations of silver, discoloration of the porcelain is obviously a consideration unless special precautions are taken during firing. Use of the alloy W1 (Williams Gold Co., Buffalo, NY) along with the low sodium content porcelain specified by the same manufacturer can eliminate this problem. Another example of a palladium–silver alloy is Silhouette 150 (Leach and Dillon, N. Attleboro, MA).

II. **Base metal alloys**
   i. ***Nickel–chromium alloy*** is the most commonly used base metal alloy and typically contains about 65%

Table 6.13 Comparison of multiple alloys according to different variables

	High Au	Low Au	Ag-Pd	Base
Composition	85% Au, 7% Pt	58% Au, 38% Pd	60% Pd, 38% Ag	65% Ni, 20% Cr
Strength (mm)	0.5	0.3	0.3	0.1
Fit	—	++	+	—
Bond	++	++	+	—
Biocompatibility	++	++	+	—
Cost	INR 6400	INR 3200	INR1600	INR 375



Figure 6.37 Porcelain-fused-to-metal fixed dental prosthesis.

nickel for strength, 20% chromium for passivity and 2% beryllium for castability and control of oxide formation. Commercial products representing this type of alloy include Rexillium III (Jeneric Pentron) and Lite-Cast B (Williams Gold Co., Buffalo, NY) (Fig. 6.37).

Non-beryllium alloys are commercially available but should not be considered for clinical use. Note that beryllium is critical to help control oxide formation with base metal alloys; however, it has been associated with the development of berylliosis, a serious occupational pulmonary condition, and laboratory technicians are at risk.

ii. ***Chromium–cobalt alloys*** are commercially available and primarily marketed as 'biocompatible' base metal alloys because they are *nickel and beryllium free*. Other than the exclusion of these elements, these alloys have little to recommend them and thus *should not be used*. Because they contain approximately twice as much cobalt as they do chromium, it has been suggested that these metals should be designated cobalt–chromium alloys. It is now possible to very accurately mill chrome–cobalt frameworks, which removes one of the major disadvantages (poor fit) of these alloys. There is little evidence at this time that the problems associated with excess oxide formation with these alloys have been solved.

iii. ***Titanium alloys*** have become available recently, and copings are currently formed both by casting and by a combination of copy-milling and spark-gap erosion. Both of these systems must be considered experimental at this time and should be used only under strict protocol in an institutional setting. The other issue with titanium fused-to-metal is that veneering ceramics with very low COEs are required, and the esthetic results achieved with these ceramics are poor.

Thus, only nickel-chrome-beryllium alloys should be the only base metal alloy to be considered for use by the practitioner.

Strength

Strength of a porcelain-bonding alloy is of paramount importance to the long-term longevity of the restoration. Any flexure whatsoever in the metal substructure will result in catastrophic failure (fracture) of the porcelain veneer. The greater the number of units in any fixed partial denture framework, the greater is the potential for flexure. An alloy will ideally possess strength in relatively thin sections to allow for minimal tooth reduction and permit the fabrication of delicate frameworks. These substructures will then permit the creation of embrasure forms and connector designs that will allow for optimum cleansibility and the development of physiologic crown contours.

I. ***High gold alloys:*** One of the major disadvantages of the high gold alloys is the relative lack of strength possessed by these metals. This lack of strength can be compensated by bulking up the framework and the occlusogingival dimension of the connector areas. This can compromise embrasure form and produce overcontoured restorations. A minimum cross-sectional thickness of **0.5 mm** is suggested for use with these alloys. High gold alloys should only be used with single crown restorations on anterior teeth. In this situation, high gold alloys have an advantage because they do not produce a dark oxide, and intense opaque porcelains do not need to be used to block out the colour of the oxide. Because of the reduced bite force with anterior teeth compared to posterior teeth, the strength of the gold framework is adequate.

II. ***High noble and palladium-based alloys:*** They are considerably stronger than the high gold alloys, and a minimum cross-sectional thickness of only **0.3 mm** is required. This reduction of thickness required is only 0.2 mm, but it is extremely significant clinically.

III. ***Nickel-chrome-beryllium alloys:*** They are very strong and can survive with a minimal thickness of only **0.1 mm**; although, it is very difficult to work within such a tight tolerance without running the risk of perforation of the metal coping.

Fit of Castings

A most important factor in long-term success with cast restorations is achieving excellent marginal integrity. A number of factors totally independent of the alloy utilized are important in this regard.

Castability of alloy systems

Distortion varies with the composition of the alloy. *The higher the melting temperature of an alloy, the less distortion will likely occur.*

- Thus, the high gold alloys, which have melting temperatures closest to the sintering temperatures of the porcelain, tend to distort the most.

- High noble and the palladium-based alloys distort somewhat less.
- Base metal alloys tend to distort very little. However, this latter fact should not be taken to construe that there is no change in fit of base metal castings during porcelain application. In fact, the quality of the fit is diminished considerably with base metal castings as a result of inevitable ***oxide*** formation on the internal of the castings. This oxide layer, which is substantial with base metal alloys, can interfere with seating of the casting and result in marginal opening.

Variations were found due to the different specific gravities of the alloys and due to variations in melting temperatures. However, current research has established that there is a *metal-mould equilibrium* for each alloy and investment combination that will produce an optimum casting in terms of completeness and the quality of the surface of the casting. *This equilibrium is defined as the difference between the melting temperature of the alloy and the temperature of the mould into which the alloy is cast*. Once this optimum temperature differential is established, excellent castings can be achieved with all of the systems under discussion. This finding, plus the development of induction casting machines, has determined castability is no longer a concern with any of the alloy systems.

Casting Shrinkage

When metal alloys are cast they solidify from the liquid to the solid state, and then subsequently cool to room temperature. Casting shrinkage occurs during this time. This shrinkage occurs in three stages:

Stage 1: When the liquid metal cools from the melting temperature to the solidification temperature
Stage 2: When it undergoes the transformation from the liquid to the solid state
Stage 3: When the solidified metal cools to room temperature. It is likely that this third stage is where the most significant shrinkage occurs

There are major differences in the amount of casting shrinkage that occurs with the various alloy systems. Casting shrinkage is a function of the melting temperature: the higher the melting temperature of the alloy, the greater will be the casting shrinkage. This shrinkage of course must be compensated by equivalent expansion of the investment. There is some question that this can be accomplished with adequate precision with the alloys possessing the highest melting temperatures. This is especially a problem with large multiunit castings.

- Type III gold alloys shrink approximately **1.2%**. Gypsum bonded investments can compensate very accurately for this amount of shrinkage.
- High gold porcelain bonding alloys shrink slightly more at about **1.3%**.
- The casting shrinkage of the high noble and palladium-based alloy systems is slightly higher than that of the high gold alloys. Compensation for this shrinkage is accomplished with high heat phosphate bonded investments, using specific amounts of liquid silica to control the expansion. The more silica used, the greater is the expansion.
- The nickel-chrome-beryllium alloys shrink **2.4%,** which is double the shrinkage of a Type III gold alloy. Early studies with these alloy systems indicated that the castings were too tight and failed to seat adequately. This was simply a result of lack of adequate expansion provided by the investment materials utilized in the investigation. Today, it is possible to provide adequate expansion with modern investments. However, the difficulty of producing an accurate casting increases as the expansions and contractions of the materials involved increase. This is particularly true when dealing with castings with non-uniform amounts of bulk, and large multiunit castings. It is difficult to solder base metal alloy units, and as a result, fixed partial dentures fabricated with this alloy are often cast in one piece. A diminution in the quality in the fit of the castings is likely to occur when this is done.

Distortion of the metal coping during firing of the porcelain veneer

Clinicians have long noted subjectively that metal copings that fit very well prior to firing the porcelain do not fit nearly as well after firing. The distortion that occurs is a function of both the composition of the alloy and the margin geometry utilized. Additionally, research has shown that not only do individual copings distort, but so do entire frameworks for larger fixed partial dentures. Because the majority of the distortion seems to occur during the *degassing process* the major factor in this latter type of distortion seems to be release of stresses induced in the casting process.

Distortion can be minimized with the intelligent use of cervical marginal geometry. *Shoulder* and *shoulder-bevel margins* distort relatively little, even with high gold alloys, and should be utilized routinely in non-esthetic situations. Framework distortion can be minimized by proper alloy selection, use of custom sager trays and use of the post-soldering procedure where indicated.

When all factors are taken into account, it would appear that the *high noble/palladium–silver/high palladium alloys* would have an advantage over the other systems in terms of the ability to achieve consistently excellent marginal integrity.

Bonding to Porcelain

Clearly the formation of a predictable bond to porcelain is an essential property of an alloy intended for crown and bridge prosthodontics. There are three main mechanisms of porcelain bonding: *mechanical*, *chemical* and *compression bonding*. The most important of these by far is chemical bonding. However, it is prudent to take advantage of all three methods.

1. ***Mechanical bonding*** occurs due to the inherent microscopically rough metal surface that exists after the degassing procedure. Van der Waal forces play a

role here, as does the improved wetting that occurs when the surface of the alloy is air-abraded with 50-μ aluminium-oxide particles prior to applying the opaque porcelain. It has been estimated that such mechanical bonding should constitute only about 10% of the total bond of the porcelain to the metal.

II. ***Compression bonding*** occurs as a result of the slight thermal COE mismatch that exists between the porcelain and a compatible alloy. When a metal-ceramic restoration is taken from the muffle of a porcelain oven and allowed to cool, the metal coping will cool first and begin to shrink slightly. This will pull the overlying porcelain under compression. This compression stress contributes to the overall bond strength. We take advantage of this by attempting to get a 'wrap around' effect with our cutback design, both buccolingually and interproximally.

III. ***Chemical bonding*** is the most important means of bonding porcelain to metal. The opaque porcelains are specially formulated with tetravalent oxides that will bond to oxides formed on the surface of the metal. The metallic oxides either form naturally or are induced from trace elements during the degas cycle, and ideally will form a monomolecular layer on the surface of the alloy (Fig. 6.38).



Figure 6.38 (A) Noble metal alloy copings. (B) Opaque porcelain applied to noble metal alloy copings. (C) Finalized porcelain-fused-to-metal crowns.

Chemical bonding with noble metal alloys: This has been extensively studied and is very predictable. Trace elements such as indium, zinc and tin are added to the alloy to provide the oxide required as well as refine the grain structure. The metal coping is run through degas or oxidation cycle to induce the oxide onto the surface. At least *50% new metal* is used each time a coping is cast to prevent depletion of the trace elements from the mix. Light air abrading with aluminium-oxide is recommended with many alloys to prevent excess oxide formation.

Chemical bonding with the base metal alloys: This is much less predictable as the problem with these alloys is with the *oxide layer*. There is simply too much of an oxide layer, and while the bond failure may occur at either the interface between the metal and the oxide or between the oxide and the opaque, it most frequently occurs within the thick oxide layer itself. *The main function of beryllium in a base metal alloy is to help control the oxide layer in addition to improving the castability of the alloy*. In the former function, the beryllium is only partially successful. Non-beryllium base metal alloys have no control of the oxide layer and hence should be avoided.

Biocompatibility

While patients can be theoretically allergic to almost any element in a casting alloy, even gold, the major elements that are of interest in terms of biocompatibility are *nickel* and *beryllium*. It is estimated that approximately 22% of women are allergic to nickel. The incidence of nickel allergy in men is less than 10%. The major reason for the difference in incidence between the sexes is thought to be that many women have pierced ears and wear costume jewellery. Much of this jewellery is nickel-based, and it is believed that this will induce the allergy.

The signs and symptoms of nickel allergy can be local or systemic. Often the presenting symptom is a rash or eczema on the arm or legs. Patients often do not associate such peripheral symptoms with an intraoral restoration and hence may suffer for prolonged periods before a diagnosis of nickel allergy is made. When use of a base metal alloy is contemplated, the patient should be specifically asked if he or she has any allergy to nickel or reaction to any metal.

Beryllium is primarily a potential problem for the laboratory technician where grindings containing beryllium can be the etiologic agent for a number of respiratory ailments. Proper ventilation and the routine wearing of a face mask can prevent such untoward events. Beryllium has also been shown to migrate toward the surface of restorations, and in that location can also dissolve in oral fluids. The clinical significance of these findings is unknown at this time but is a concern to some.

Economic Considerations

Obviously, the cost of a metal alloy needs to be considered. This can be even more important when the price of gold fluctuates extensively. Indeed the genesis of most of the research

into alternative alloys, and many of the all-ceramic alternatives to metal-ceramics, was a gold price ranging $1000 to $1600 an ounce. Indeed, a major impetus for the development of all-ceramic alternatives to PFM is the laboratory costs.

In summary, a host of metal alloys intended to be used with porcelain are currently available commercially. These have been classified according to both composition and the characteristics and physical properties of the major groups compared.

An analysis of the material presented reveals that one group of alloys, the ***high noble alloys***, seems to have no deficiencies and scores high in all of the categories discussed. High noble alloys also happen to be cheaper than high gold alloys, although that is a secondary consideration. These alloys would seem to be the *metals of choice*. They are stronger and can be used in thinner section than the high gold alloys. They cast easily and accurately, and show fewer tendencies to distort on firing the porcelain. Porcelain bonding is very predictable with no discolouring of the porcelain. Biocompatibility is not a problem.

If cost was the major consideration, use of a palladium silver alloy with a porcelain low in sodium content would likely be a suitable choice. As is usually the case, proper manipulation of the alloy chosen is probably even more important than which alloy is chosen.

Clinical Notes

I. Porcelain-fused-to-metal crowns and fixed partial dentures continue to be important restorations, and provide the best combination of reasonable esthetics with maximum longevity of available esthetic crown options.
II. Many different alloy systems with different properties and costs are available.
III. The dentist should choose which alloy is most appropriate for each restoration.
IV. Some of the factors that must be considered when choosing an alloy include strength, ability to achieve excellent marginal integrity, ability to attain a predictable bond with porcelain, biocompatibility and cost.
V. Base metal alloys are inexpensive, but have significant problems with fit, porcelain bonding and biocompatibility.
VI. High gold alloys do not form dark oxides, and have the potential to provide improved esthetics for single crowns on anterior teeth.
VII. High noble alloys are less expensive than high gold alloys, and seem to be the alloy of choice for metal-ceramic restorations.

Evolution of All-Ceramic Crowns

New ceramic systems continue to be introduced to the profession at exponential rates. It is critical that clinicians understand how these systems have evolved since the mid-1980s and they also understand the deficiencies of many of these systems so they can more effectively evaluate the potential of new systems as they are introduced to the market. The three major concerns with new systems are as follows:

I. What is the real esthetic potential of the system compared to available tried and true systems?
II. What is a reasonable potential life span for the new system? This can be difficult to determine because improvements in physical properties do not necessarily translate into improved longevity. Keep in mind that ceramics do not fail because they are not strong enough; they fail because it is not known how to manufacture flaw-free restorations. Those flaws, called *Griffith flaws*, tend to propagate over time and result in fatigue failure.
III. What clinical situations are recommended for the new system? Can fixed partial dentures made with this system reasonably be expected to survive? Should its use be limited to anterior teeth or can it be used in molars?

This section has been written to describe the main features of the different ceramic systems, as they were introduced to the profession (Fig. 6.39). It is not our intent to examine each system in great detail, but to highlight the important features that led to success or failure. It is also not intended to be all-inclusive, but will focus on the major players.

One excellent systematic review published in 1989 indicated the failure rate with PFMs is 5% at 11 years.[38] Several other studies have shown similar results.[39] Such data must be considered when assessing the utility of new ceramic systems. Numerous ceramic systems have been introduced to the profession. Almost all of these systems have been brought to market with almost no clinical testing. It is also important to repeat the fact that laboratory testing of physical and mechanical properties and load-to-failure testing yields little useful information about expected clinical performance. Thus, it is clear that clinical dentists are carrying out the essential clinical trials for the manufacturers.

Porcelain Jacket crown (PJC): The first esthetic crown was the *PJC* introduced to the profession circa 1910 by Land. To fabricate this crown, a platinum foil was adapted to the die using a tinner's joint. The PJC had good esthetics, but poor marginal integrity and a short life span. They were cemented in place using different shades of silicophosphate cement. The major deficiency of the PJC was the fact they frequently fractured 1 or 2 years after placement.

In 1962, Weinstein-Katz and Weinstein filed the patent for the first metal-ceramic crown system.[37] At the time, a

1. Feldspathic PJC
2. Aluminous Core PJC
3. Foil Reinforced
4. Captek
5. Cerestore
6. Dicor
7. High-Ceram
8. Willi's Glass
9. IPS Empress
10. Optec
11. Empress II
12. IPS Eris
13. IPS e.Max
14. Inceram (spinell, alumina, zirconia)
15. Procera All-Ceram
16. Zirconia cored materials

Figure 6.39 Major ceramic systems used in restorative dentistry.

porcelain-fused-to-metal (PFM) was known as a ***Ceramco***. It became popular in the late 1960s, but dentists knew little about the important preparation details required to produce a truly esthetic restoration. Dentists did not understand the space requirements needed for layers of metal, opaque porcelain, body and incisal porcelains and thus most preparations were underprepared. They knew nothing about cervical margin design. Laboratory technicians were also unaware of many of the important details of fabrication. The dental profession and laboratory industry learned about PFMs by trial and error. This resulted in many restorations that lacked the expected esthetic potential and frequent fracturing of the feldspathic porcelain veneer from the metal coping.

Aluminous core PJC: In 1978, McLean introduced the *aluminous core PJC*.[40] This restoration used twin platinum foils for fabrication. One foil was used as a die spacer and the other was tin plated and used to form the core with aluminous core porcelain. This was veneered with compatible low-alkali porcelain. The laboratory technique was very demanding, which limited its commercial adaptation.

Foil reinforced: In 1984, Donovan introduced a simplified single foil technique for fabricating aluminous core PJCs, but this simplified technique also failed to catch on.[41] The twin-foil crown by McLean was the first in a series of foil-supported restorations that attempted to improve clinical longevity by reducing the number of microscopic defects on the internal portion of the crown.

Shorer introduced the Renaissance foil system, which used a pleated noble metal foil and a gold coloured bonding agent. Asami Tanaka had a similar system with his Sunrise crown. These foil-supported restorations had poor survival data, primarily because the foil took the form of the tooth preparation and frequently there were excesses of unsupported porcelain. The industry standard is that there should never be more than 2 mm of unsupported porcelain, regardless of the core material. With foil-supported crowns, this is difficult to achieve in the majority of cases.

Captek system: In 1993, Shorer and Whiteman introduced the Captek system. Captek is an interesting technology that forms a noble metal coping without casting or machining. A wax sheet containing a powdered metal alloy of 88% gold is adapted to a refractory die and heated in an oven until the wax melts and the metal coalesces into a porous coping. A second wax sheet containing platinum is adapted over the coping and heated. The wax melts and the platinum enters the gold coping by capillary action (hence, Captek). The gold colour of the coping results in excellent esthetics and the platinum provides strength to resist metal flexure. The major deficiency of Captek is the inability to provide adequate support for the veneering ceramic, again because the coping takes on the form of the tooth preparation and few preparations are ideal.

Cerestore: In 1983, Sozio and Riley introduced the Cerestore crown, also called the 'shrink-free' ceramic crown.[42] This product used an extremely accurate epoxy-resin die system, on which was injected a high strength alumina-ceramic core. This core did not shrink on sintering and was quite opaque, which resulted in unesthetic cervical opacity. The core could be coloured intrinsically and was overlayed with a relatively translucent veneering ceramic. The Cerestore crown system had a relatively short time on the market, partially due to the inherent cervical opacity which resulted in limiting the esthetic outcome to that attained with metal-ceramic restorations with porcelain labial margins, and partially due to early clinical failures with fracture.

Experiences with Cerestore taught us one important principle: *If an all-ceramic crown system uses an opaque core to improve fracture strength, the esthetic result achieved by that system will be no better than the esthetics achieved with metal-ceramics.*

Dicor: About the same time as Cerestore entered the marketplace, Dicor was introduced and marketed as the 'castable' ceramic crown.[43] Dicor crowns were waxed, invested and cast using a glasslike material that was very translucent. The restoration was then given a specific heat treatment called 'ceraming', which induced a change in crystalline structure and made it slightly less translucent. Then a thin film of extrinsic stain was painted on the crown to attain the desired shade.

Dicor was the first example of a monolithic crown system where the final esthetic result was attained by applying a thin layer of extrinsic coloration (Fig. 6.40). In general, such monolithic restorations lack depth and have a high level of surface or specular reflection. Such restorations often experience a high incidence of metamerism where the restoration appears one colour with certain light sources and a very different colour in different light sources. Thus, the major deficiency of Dicor was the inability to achieve intrinsic coloration. Fracture rates for Dicor crowns were also relatively high.

Originally, Dicor crowns were intended to be cemented with different shades of zinc-phosphate cement. However, Malament discovered that fracture rates could be improved if Dicor crowns were etched internally and bonded in place with resin cements. The combination of high fracture rates and poor esthetics led to the early demise of Dicor, however.

Experiences with Dicor taught us two important principles: *First, monolithic crowns do not allow intrinsic coloration and provide depth of colour in the same manner as layered crowns; second, the clinical performance of ceramic crowns can be improved by etching and bonding.*



Figure 6.40 Ceramic restoration by Dicor.



Figure 6.41 Leucite-reinforced glass ceramic crowns (IPS Empress).

Willi Glass crown: To overcome the esthetic disadvantages of Dicor, Willi Geller introduced the Willi Glass crown in 1987.[44] Willi Glass used an internal translucent core of Dicor, about 0.75 mm thick. This was veneered with Vitadur N ceramic. Willi Glass was undoubtedly the most esthetic restoration ever made available to the profession. However, it had a relatively short clinical life span and suffered from catastrophic fracture. This was partially because the Dicor core was not strong enough at that thickness, and partially due to thermal incompatibilities of the Dicor core and the veneering porcelain.

IPS Empress: In the late 1980s, Wholwend and Scharer introduced the original IPS Empress leucite reinforced ceramic system (Fig. 6.41).[45] This system introduced the concept of hot pressing the ceramic, which has the potential to reduce the number of microscopic defects in the ceramic crown.

Leucite-reinforced ceramic systems similar to IPS Empress (Optec HSP, Authentic) have appeared on the market. Optec HSP was a spectacularly unsuccessful product, which illustrates another important principle: *Every ceramic system, even if similar to another successful system, must generate its own clinical data to support use of the system.*

In-Ceram: In the early 1990s, Vita introduced the In-Ceram system, utilizing slip-casting with infiltrated glass. Three different forms exist:

I. In-Ceram Spinell: Spinell is the most translucent but has not performed well in clinical trials.
II. In-Ceram Alumina: In-Ceram Alumina has decent esthetic potential and has been used successfully for single crowns.
III. In-Ceram Zirconia: Zirconia is the least translucent and has the same translucency as metal; it therefore has limited esthetic potential.

These systems have not captured a major share of the market to date.

Procera Allceram: Nobel Biocare introduced Procera Allceram to the market in the 1990s. This system used a milled alumina or zirconia core veneered with low fusing ceramic. Crowns made with this system had inferior marginal integrity and a high incidence of chipping of the veneering ceramic because of lack of support of the ceramic by the core.

IPS Empress II: IPS Empress II was introduced as a lithium disilicate reinforced material that was so strong it could be predictably used to fabricate three-unit fixed partial dentures. In this regard, IPS Empress II was a complete disaster. One local dental laboratory (Drake Dental Lab, Charlotte, NC) fabricated 360 IPS Empress II fixed partial dentures using Ivoclar's specification for connector height, and at 3 years, the failure rate due to fracture was 75% (personal communication; David Avery, Drake Dental Laboratory).

It should be noted that IPS Empress II was a lithium disilicate material and was not really related to the original IPS Empress, which was leucite reinforced. Part of the problem with IPS Empress II was that the veneering ceramic was not properly matched to the lithium disilicate core. This was corrected in a later version called IPS Eris. IPS Eris has shown promise for single crowns, but should ***not*** be used for fabricating all-ceramic fixed partial dentures. IPS Eris has largely been made irrelevant with the introduction of IPS e.max, which is an optimized lithium disilicate material.

IPS e.max: IPS e.max is a lithium disilicate material that was introduced in 2009 (Fig. 6.42). It can be pressed or machined and can be used as a monolithic material or layered. Monolithic IPS e.max has relatively high flexural strength of 360–400 MPa compared to IPS Empress, which has a flexural strength of approximately 160 MPa. Layered IPS e.max has approximately half the flexural strength as the monolithic form. No long-term clinical trials have been published at this point, but information gleaned from remake data from three commercial laboratories and over 21,000 IPS e.max monolithic and layered single crowns indicates the fracture rate at about 4 years is only 1%. As would be expected, *layered IPS e.max crowns have better esthetic potential than monolithic crowns, but the monolithic crowns are stronger.*

It should again be noted that higher strength does not automatically result in improved clinical performance, primarily because all-ceramic crowns do not fail because they are not strong enough. They fail because it is difficult to fabricate them without incorporating microscopic defects called *Griffith flaws*, and it is the propagation of these defects that results in catastrophic failure.



Figure 6.42 Lithium disilicate (IPS e.max) ceramic fixed dental prosthesis.

Zirconia: In the late 1990s, zirconia-based materials were introduced to the profession. The original restorations were cored with yttria-stabilized zirconia oxide and veneered with a compatible ceramic. Zirconia is a very strong material with flexural strengths as high as 1400 MPa. In clinical trials with these materials, core fracture was extremely rare even with posterior fixed partial dentures. However, the 'Achilles heel' with these layered restorations was chip fracture of the veneer. The fractures did not occur at the interface between the core and the veneer, but rather were cohesive fractures in the body of the ceramic veneer. One clinical trial of over 900 crowns estimated that layered zirconia crowns fractured at a rate 5 times higher than that of metal-ceramic crowns.

There are two potential causes for this high incidence of veneer fracture.

I. Because the core is computer designed and manufactured from a scan of the tooth preparation, it is highly likely that an adequate support of the veneering porcelain is not provided by the core. This problem may have been solved by creating software that produces a virtual full-contour wax-up and then cuts that back to provide a core with optimal support.
II. The second cause is the extremely poor thermal conductivity of the core material. Basically the core cools very slowly on removal from the porcelain oven and the veneering ceramic cools past its glass transition temperature while the core slowly continues to cool. This creates stress in the veneering ceramic, which causes it to fracture under occlusal loads. This problem may have been solved by using prolonged cooling times.

The reason core fracture is so rare with zirconia is that zirconia possesses a property called transformation toughening. When a defect begins to propagate, the material surrounding the defect undergoes a transformation from the tetragonal to the monoclinic phase, and this transformation is accompanied by a slight expansion, which effectively pinches off the defect and prevents it from propagating.

A relatively new zirconia restoration has been introduced in recent years. It is called ***monolithic zirconia*** or ***full contour zirconia*** (Fig. 6.43). This is basically a core of zirconia milled into a full crown and then either glazed or polished. Based on studies of layered zirconia, where core fracture is extremely rare, these restorations may reasonably be expected to have high survival rates. One major advantage of monolithic zirconia is that very conservative tooth preparations can be used because of the strength of the material. On molars, preparations similar to those used for cast gold crowns will suffice.

Full contour zirconia restorations are also relatively inexpensive. This is partially because they are designed and manufactured digitally. In an era where the price of gold is over or close to $1600/ounce, this low cost benefit is a real advantage.

There is no question that the esthetic results achieved with these restorations are compromised, but they are acceptable for molars and perhaps mandibular incisors.



Figure 6.43 Monolithic full contour zirconia restoration.

Research has shown that the best surface for full contour zirconia crowns is a polished surface. This polished surface is relatively kind to opposing enamel.

The preceding information is not meant to be either all-inclusive or a detailed description of available ceramic systems. Rather, it is an overview of how these systems evolved with a brief description of the important features of the major players.

Clinical Notes

I. Many different ceramic systems have been introduced to the dental profession over the past 30 years.
II. All have been brought to market with little or no clinical testing.
III. Ceramic materials do not fracture because of a lack of strength. They fracture due to defect propagation and fatigue.
IV. Because of #III, physical properties testing and load-to-failure studies yield little useful information on potential clinical survival rates.
V. Clinical trials or fatigue testing under water are the only appropriate means of predicting clinical survival of ceramic crowns.
VI. Most of the ceramic systems introduced since 1980 have been clinical failures.
VII. The current cost of gold is a major factor in the cost of restoring teeth with metal-ceramics.
VIII. Many all-ceramic systems, especially full contour zirconia, are cost-effective compared to PFM.
IX. Monolithic crowns have good strength but relatively poor esthetics.
X. Layered restorations have good esthetic potential but reduced strength.
XI. Current available systems that clinicians should consider are:
    i. Metal-ceramics: Reasonable esthetics, longest life span
    ii. IPS Empress: Esthetics, anterior teeth only
    iii. IPS e.max: Esthetic potential, potential longevity
    iv. Layered and full contour zirconia: Potential longevity, cost

Contemporary Ceramic Materials

There are a number of reasons why dentists fail to obtain predictable esthetics with metal-ceramic restorations and most of them are *operator-related*.

I. Most dentists tend to underprepare teeth that are to be restored with metal-ceramic crowns, and do not provide the laboratory technician sufficient room for the metal substructure, opaque layer and body and incisal porcelain.
II. Many use outdated cervical margin designs, and soft tissue management is generally poor, resulting in a high incidence of inadequate impressions.[46,47]
III. Many dentists opt to send their laboratory work to substandard laboratories and accept low-quality restorations in the name of cost reduction.

It is clear that with proper tooth preparation and soft tissue management, coupled with working with a decent ceramist, very esthetic restorations can be fabricated using metal-ceramics. Metal-ceramic restorations remain the gold standard for esthetic indirect crown systems because they provide a combination of reasonable esthetics with maximum longevity. Data from a number of clinical trials indicate that *95% of metal-ceramic restorations are in service 10–15 years after placement*.[38]

Since the early 1980s, many different ceramic systems have been brought to market. None of the systems had advanced clinical testing before coming to market and most systems failed primarily due to premature catastrophic fracture. The driving force behind these new systems was twofold:

I. Many clinicians were disappointed with the esthetic results they obtained with metal-ceramic restorations and hoped to obtain better esthetic results by using metal-free restorations.
II. The second driving factor was the cost of noble metals. In 2016, the cost of gold ranged from $1200 to $1600 an ounce. This resulted in skyrocketing laboratory costs in a market that was not favourable to raising compensatory fees. Some of the newer systems offer significant savings in laboratory costs, partially due to nonuse of metal but also because many of these restorations can be fabricated digitally thus reducing labour costs.

There are three major materials that have captured a significant portion of the ceramic market in 2017 that have not had properly conducted clinical trials published before they were widely used clinically. These three materials include:

I. Leucite-reinforced IPS Empress (Ivoclar)
II. Lithium disilicate–reinforced IPS e.max (Ivoclar)
III. Zirconia-based systems (many manufacturers)

All of these materials can be used in the monolithic state or layered. Zirconia materials can be used layered in the partially stabilized state and monolithic in both the partially stabilized and fully stabilized state.

Monolithic ceramic restorations have relatively poor esthetic results, but seem to provide longer clinical survival rates. With monolithic ceramic restorations, all of the colour is on the surface, resulting in high surface reflectivity and specular reflection, a high incidence of metamerism and no depth effect. Monolithic crowns are *not* suitable for maxillary anterior teeth. Layered restorations provide significantly improved esthetic results but lower survival rates.

I. ***Leucite-reinforced IPS Empress system***: This was really the first successful ceramic system on the market appearing after an imposing list of failed systems.
   i. Layered IPS Empress should only be used on anterior teeth as clinical trials have demonstrated high failure rates on posterior teeth.[48]
   ii. Monolithic Empress should not be used.
   iii. IPS Empress can be pressed or milled, and the usual procedure is to press a leucite core and veneer it with the matched ceramic.

   IPS Empress crowns can be fabricated in two different ways.
   i. *Monolithic IPS Empress* crowns can be waxed, pressed and stained. These monolithic restorations suffer from issues discussed previously and do not provide optimum esthetics on anterior teeth.
   ii. *Layered IPS Empress*, with a leucite reinforced core and matched veneering porcelain, *provides the most esthetic and durable ceramic option currently available for restoring anterior teeth*. Clinical trials have indicated 95% survival of bonded, layered IPS Empress crowns on anterior teeth. Survival rates drop considerably with crowns on posterior teeth.

   It is critical to understand that this high survival for IPS Empress crowns on anterior teeth is only achieved if they are internally etched and bonded in place. There are three approaches to accomplishing this.
   i. The first is the traditional approach of bonding the restoration in place using a 4th or 5th generation etch-and-rinse dentin bonding agent (DBA) and a dual cure resin cement. One problem with this approach is if the DBA is cured prior to cementation, the thickness of the cured DBA may prevent complete seating of the crown. Alternatively, if the DBA is cured after seating the crown at the same time as the dual cure cement, the hydraulic pressure generated during seating will likely collapse the collagen bundles and significantly compromise the hybrid layer and dentin bonding.
   ii. A second approach is to use Paul and Magne's immediate dentin sealing technique.[49] With this technique the dentin is sealed with a 4th-generation DBA immediately ***after*** tooth preparation, prior to making the impression. Thus, the film thickness of the DBA is of no importance, and the tooth is sealed during the provisional phase. The sealed preparation must be isolated with Vaseline when making

Table 6.14 Mechanical properties of leucite reinforced, lithium disilicate and zirconia

Materials	Flexural strength (MPa)	Elastic modulus (GPa)	Hardness (VHN)
Leucite reinforced glass ceramic (Empress)	160	96	550
Lithium disilicate (IPS e.max)	400	95	5800
Zirconia (Y-TZP)	800–1500	210	1200

the provisional restoration to prevent bonding of the provisional to the dentin. The oxygen-inhibited layer of the DBA can also distort impressions and the surface of the DBA must be carefully cleaned prior to cementation. This immediate dentin sealing technique results in much higher dentin bonds than are created when bonding at the cementation stage.

iii. The third approach to bonding etched IPS Empress restorations in place is to use self-adhesive resin cement. These cements do not produce a hybrid layer and strong bonds to dentin, but do provide an excellent seal. The long-term effect on crown survival with use of self-etching cements is not known.

Fewer and fewer dental laboratories are fabricating these restorations today, and even though the esthetic results are excellent and survival on maxillary anterior teeth is satisfactory, their use in contemporary clinical dentistry is limited.

II. ***Lithium disilicate–reinforced IPS e.max***: The material named IPS e.max is a lithium disilicate reinforced ceramic that can be used both in the monolithic state and layered.

i. Most of what is known about the clinical performance with IPS e.max is with monolithic restorations.

ii. Monolithic e.max has a significantly higher flexural strength than IPS Empress at 400 MPa versus 160 MPa (Table 6.14).

iii. The flexural strength of layered e.max is about half of the monolithic form which has been validated with both fatigue studies and clinical data.[50,51] It is important to remember that ceramic crowns fail by defect propagation, and it is likely that the clinical success of e.max restorations is a result of incorporating fewer inherent defects through pressing and milling the restorations. As impressive as the gain in flexural strength of monolithic e.max versus IPS Empress is, it pales in comparison to the flexural strength of monolithic zirconia, which has a flexural strength of 1500 MPa versus 400 MPa for e.max.

III. ***Zirconia-based systems***: Zirconia restorations were first introduced to the profession as partially stabilized (yttria) zirconia oxide cores layered with a veneering ceramic.

i. *Layered Zirconia*: Initial clinical data established that the cores rarely fractured, but there was a high incidence of chip fractures of the veneering ceramic.[52] It was estimated that the incidence of veneer fracture with layered zirconia was 5 times that of metal-ceramic restorations. The low incidence of core fracture could be attributed to the property of *transformation toughening* (Fig. 6.44). When a defect begins

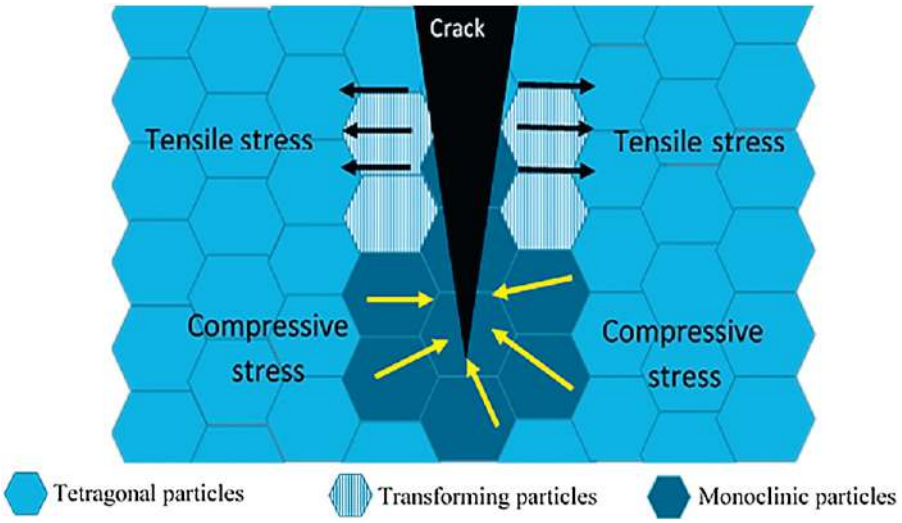


Figure 6.44 Schematic illustration of the transformation-toughening phenomenon.

to propagate in a zirconia core, the materials immediately surrounding the defect change from the tetragonal to the monoclinic form of zirconia. This results in a slight expansion, which effectively 'pinches off' the propagating defect.

– Veneer chipping is likely caused by two issues:
  - First, the high strength cores were originally fabricated by scanning the die of the tooth preparation and then a core of uniform thickness was milled. This core would follow the shape of the die and does not in any way account for the thickness of the veneering ceramic. The industry standard for support of the veneering ceramic is that the veneering ceramic should never exceed a thickness of 2 mm. New software that first produces a virtual full-contour image of the anticipated final restoration and then cuts back that restoration, to provide a core with optimum support of the veneering ceramic, should solve the support problem.
  - The second issue with veneer chipping is the thermal conductivity of zirconia. It is very low, perhaps 100–200 times lower than the conductivity of metals used to bond to porcelain (Table 6.15).
– The net result of this is, as the layered ceramic unit exits the porcelain oven, the veneering ceramic cools first, below its glass transition temperature while the core very slowly continues to cool and shrink. This creates stress in the veneering ceramic, which is expressed clinically some time later by chip fracture. Slower cooling protocols have been introduced for firing layered zirconia restorations, and it is anticipated this will significantly reduce the incidence of chipping.[7]

ii. *Partially stabilized Monolithic zirconia* (*also called full contour zirconia*): A relatively new form of zirconia has become extremely popular in recent years despite a complete lack of clinical data on survival. Monolithic partially stabilized zirconia (also called full contour zirconia) has grabbed a huge share of the world market for indirect esthetic crowns. The rationale for this restoration is that in clinical trials of layered zirconia restorations core fracture is a relatively rare occurrence. A monolithic zirconia crown is essentially a core with a glazed or polished surface, and thus is expected to display good survival rates.

– One major advantage of partially stabilized monolithic zirconia is that crowns can be fabricated with significantly more conservative tooth preparations compared to those needed for metal-ceramic and lithium disilicate crowns. The exact tooth reduction required is unknown at this time, but a 1-mm occlusal reduction with axial reduction of 0.75 mm and a well-defined chamfer margin seem appropriate.
– A second advantage is that numerous fatigue wear studies indicate that polished zirconia is extremely kind to opposing tooth structure.
– A third and apparently compelling advantage of these restorations is that they are very inexpensive.

One significant disadvantage related to partially stabilized monolithic zirconia is that it is *relatively opaque*. The colour can be improved using glazes and external colorants, but the overall esthetic result is mediocre at best.

The esthetic results are likely acceptable in the molar regions for all but the most demanding of patients, but they are likely contraindicated for routine use in the maxillary anterior area.

iii. *Fully stabilized monolithic zirconia*: The relative opacity has resulted in the introduction of a new zirconia-based system: fully stabilized monolithic zirconia (Bruxzir Anterior, Glidewell Laboratory). These materials consist of zirconia oxide fully stabilized with yttria (8–12 wt%). They also contain cubic zirconia of up to 53%, and are slightly less opaque than partially stabilized zirconia. This improvement in opacity is likely not within the realm of human perception and is accompanied by a 50% reduction in flexural strength. Clinicians are advised to adopt this new technology with extreme caution. There is no clinical track record, improvements in translucency are minimal, and the reduction in flexural strength is significant (Fig. 6.45).

Table 6.15 Thermal conductivity of zirconia compared to other material

Materials	Thermal conductivity W/(mK)
Silver	429
Gold	318
Gold alloy	200
Base metal	40
Alumina	30
Zirconia	2

A Note on Survival Rates

In the absence of properly conducted clinical trials on the previously discussed ceramic systems, another approach has been used to generate timely clinical data on the survival of these restorations. Dental laboratories often warranty their crowns for 5 years and have meticulous data on remakes with these new ceramic systems. Assuming that most patients would return to their original dentists when experiencing premature failure with a ceramic crown to take advantage of the warranty, and assuming the dentists will return that restoration for remake to the same laboratory, examination of remake data provides insights of



Figure 6.45 Monolithic zirconia fixed dental prosthesis. (Case courtesy Dr.Vihelm Olaffson.)

short-term to medium-term performance of large numbers of ceramic crowns.

This 'laboratory survey' approach has provided 4-year data on over 21,000 lithium disilicate (e.max) restorations and 5-year data for over 71,000 monolithic and layered zirconia (partially stabilized) restorations.[53,54]

- Monolithic e.max single crowns fractured at a rate of 0.91% at 4 years
- Layered e.max single crowns exhibiting double the fracture rate at 1.81%
- Monolithic zirconia single crowns had an even lower rate of fracture (0.71%) at 5 years.
- Layered zirconia single crowns had a 4–5 times higher fracture rate at 3.24% at 5 years. These restorations were fabricated prior to initiation of the new protocols expected to reduce the chip fracture rate of layered zirconia restorations.
- Fracture rate of e.max inlays/onlays, which was 1.01% at 4 years.

A Note on Bonding Zirconia

- Zirconia cores cannot be effectively etched with any known acid. Zirconia restorations are best used with tooth preparations with adequate resistance and retention form, and should be cemented with *RMGI* or *self-adhesive resin cement*.
- A number of different protocols have been developed for 'bonding' zirconia restorations. All of these protocols involve air particle abrasion, use of a primer with MDP monomer and silane, and use of dual-cure resin cement. The bonds achieved with these protocols are initially weak and get weaker with artificial aging. There is a definite risk in use of air particle abrasion with zirconia restorations as some studies have demonstrated significant conversion of the core to the monoclinic form and substantial weakening after air particle abrasion.
- Zirconia seems to have a strong affinity for proteins found in saliva and blood. It is likely the internal portion of most zirconia crowns will be contaminated during try-in. These proteins cannot be removed with phosphoric acid. A routine procedure for cementing zirconia crowns is the use of a sodium hydroxide solution (Ivoclean, Ivoclar) for 20 s after try-in before cementation to effectively remove these proteins.

Indications

- *Metal-ceramic restorations* can continue to be recommended for indirect esthetic crown restorations.
- Monolithic partially stabilized zirconia crowns: It would seem reasonable to consider use of monolithic partially stabilized zirconia crowns for a molar tooth, in part due to their substantially reduced costs, and also because a more conservative tooth preparation can be used. These restorations can be recommended for mandibular anterior teeth because they are usually not highly visible and a more conservative tooth preparation can be used.
- *Monolithic lithium disilicate crowns*: They can be recommended for routine use on premolar teeth due to their excellent survival rates, adequate esthetic results, and reduced costs.
- *Layered lithium disilicate crowns*: They can be recommended for maxillary anterior teeth.
- *Fully stabilized zirconia crowns*: They should not be used at this time until clinical information is available on clinical survival.

Clinical Notes

I. Metal-ceramic restorations continue to provide an impressive combination of reasonable esthetic results coupled with excellent clinical longevity.
II. Most ceramic systems introduced to the market prior to 1988 were clinical failures.
III. Ceramic systems should be documented with independent clinical trials that demonstrate 95% survival at 5 years.
IV. The 'laboratory survey' approach is a good one for obtaining clinical short-term survival rates.
V. Monolithic zirconia and lithium disilicate materials have adequate short-term survival rates.
VI. Early results with layered zirconia showed unacceptable levels of chip fractures of the veneering ceramic. These problems may be solved with new approaches for support of the veneer and new firing cycles.
VII. Fully stabilized zirconia crowns marketed for anterior applications have minor improvements in translucency and significantly reduced flexural strength.
VIII. It is not practical with current protocols to 'bond' zirconia restorations. Zirconia restorations require traditional resistance and retention form and should be cemented with RMGI or self-adhesive resin cement.

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